Joint Distribution, independence
Markov Model
Bayesian Neworks

Chris Ding

A markov chain question
A Bayesian Network example

Joint Distribution, Independence

joint distribution P(Game, Snack) 联合分布

Sample 115 people. Their preferences on snacks and games.

Game \ Snack	Pizza Rolls	Chips and Dip	Cookies	**Totals**
Poker	10	3	12	**25**
Trivial Pursuit	8	14	7	**29**
Monopoly	14	17	7	**38**
Wii Bowling	12	7	4	**23**
Totals	**44**	**41**	**30**	**115**

joint distribution P(Game, Snack)

联合分布;

Marginal distribution 边缘分布 P(Game)

$$P(G) = \sum_S P(G, S)$$

Game \ Snack	Pizza Rolls	Chips and Dip	Cookies	**Totals**
Poker	10/115	3 /115	12 /115	**25**/115
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Monopoly	14 /115	17/115	7 /115	**38** /115
Wii Bowling	12 /115	7/115	4 /115	**23**/115
Totals	**44**/115	**41**/115	**30** /115	**115** /115

Marginal distribution P(Snack)

$$P(S) = \sum_G P(G, S)$$

Joint distribution: independence between random variables

Game and Snack are independent if $P(G,S) = P(G)P(S) \quad \forall\, G,S$

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In real life, $P(G,S) = P(G)P(S) \quad \forall\, G,S$ never happened , unless the data are man-made deliberately.

because G,S have randomness/uncertainty.

This is true even if G,S are inherently unrelated: G = person's height, S = person smoke or not.

So, how do we determine from sampling data that G and S are independent? **Chi square test** !

Computer Homework

Random Variable Independence in Probability

- N fair, independent coin flips:

$P(X_1)$

H	0.5
T	0.5

$P(X_2)$

H	0.5
T	0.5

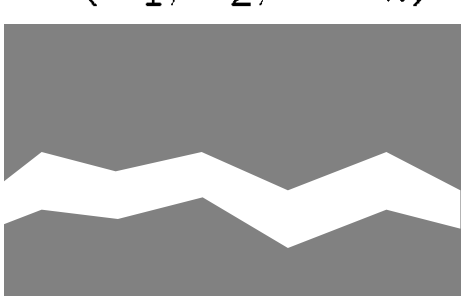
$\ldots$

$P(X_n)$

H	0.5
T	0.5

$P(X_1, X_2, \ldots X_n)$

2^n



Homework Problem

1.Get 2 coins, marked as X1 or X2.
2.Throw them N=20 times. Check heads and tails.
3.Record the results in 2x2 table.
4.Compute joint and marginal probabilities.
5.Do you get P(x,y) = P(x)*P(y) ?
6.If not Compute chi-square. Independent if < 3.84

Hint: use random to simulate coins. X1={1,0}.
Similarly for X2.

Chi-square test of independence between random variables

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Observed data

Assume Game and Snack are independent.
Then the theoretically expected value :

$$E_{ij} = NP(G_i)P(S_j)$$

$$E_{11} = NP(G_1)P(S_1) = (115)(25/115)(44/115) = 9.565$$

$$\chi^2 = \sum_{i,j}\left[\frac{(O_{ij} - E_{ij})^2}{E_{ij}}\right] = \frac{(10 - 9.565)^2}{9.565} + \cdots = 14.508.$$

At 95% confidence with (3 -1) x (4 -1) = 6 degrees of freedom, chi-square distribution value is $12.592.$

Because $\chi^2 > 12.592.$ We conclude that G and S are not independent with 95% confidence.

Deviation from independence exceeds theoretically expected bound

Chi-square test of independence between random variables

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$$\chi^2 = \sum_{i,j}\left[\frac{(O_{ij}-E_{ij})^2}{E_{ij}}\right] = \frac{(10-9.565)^2}{9.565} + \cdots = \sout{14.508.}$$

= 10.281 ← **What if**

At 95% confidence with (3 -1) x (4 -1) = 6 degrees of freedom, chi-square distribution value is $12.592.$

Because $\chi^2 > 12.592.$

~~We conclude that G and S are not independent with 95% confidence.~~

There is no evidence we can reject with 95% confidence the null hypothesis that G and S are independent.

Deviation from independence exceeds theoretically expected bound

With 95% confidence the null hypothesis that G and S are independent is consistent with data.

Suppose chi-square test show the null hypothesis that G and S are independent is consistent with data. Then we model the data using independence. This greatly simplifies the analysis.

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Marginal distribution becomes more important.
Model parameters are marginal distributions:

$$P(G_1), P(G_2), \cdots, P(G_m)$$

$$P(S_1), P(S_2), \cdots, P(S_k)$$

Joint distribution are not the model parameters:

$$\begin{pmatrix} P(G_1, S_1) & P(G_1, S_2) & \cdots & P(G_1, S_k) \\ \vdots & \vdots & \vdots & \vdots \\ P(G_m, S_1) & P(G_m, S_2) & \cdots & P(G_m, S_k) \end{pmatrix}$$

For example, we assert $P_{11} = P(G_1)P(S_1) = (\frac{25}{115})(\frac{44}{115}) = 0.08318$

$$P_{11} \neq \frac{10}{115} = 0.08696$$

How do you compute joint and marginal probabilities?
Use multinomial distribution and maximum likelihood.
Normal distributions => different results

Bayesian network use variable independence to simplify analysis

Bayesian network use variable independence to simplify theoretical analysis

Markov Model,

conditional independence

Markov Model: conditional independence







Day: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 …

Weather: S S R C C R R S S S R C C R S …

Markov assumption:
tommorrow's weather
depends only on today's weather,
independent of yesterday's weather, day before yesterday's weather, etc

$$P(X_{t+1} \mid X_t, X_{t-1}, X_{t-2}, \ldots) = P(X_{t+1} \mid X_t, \cancel{X_{t-1}}, \cancel{X_{t-2}}, \ldots)$$

Markov Model: conditional independence







Day: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 …

Weather: S S R C C R R S S S R C C R S …

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$$P(X_{t+1} \mid X_t, X_{t-1}, X_{t-2}, \ldots) = P(X_{t+1} \mid X_t, \cancel{X_{t-1}}, \cancel{X_{t-2}}, \ldots)$$

$$P(A,B,C) = P(A|B,C)P(B,C) = P(A|B,C)P(B|C)P(C)$$

$$\begin{aligned} P(X_{t+1}, X_t, X_{t-1}) &= P(X_{t+1}|X_t, X_{t-1})P(X_t, X_{t-1}) \\ &= P(X_{t+1}|X_t, X_{t-1})P(X_t|X_{t-1})P(X_{t-1}) \\ &= P(X_{t+1}|X_t, \cancel{X_{t-1}})P(X_t|X_{t-1})P(X_{t-1}) \\ &= P(X_{t+1}|X_t)P(X_t|X_{t-1})P(X_{t-1}) \end{aligned}$$

Equavalent conditions for conditional independence

Proposition (1.1)

Variables X and Y are conditionally independent given Z if and only if one of the following conditions is met:

1. $P(X|Y,Z) = P(X|Z)$ *if* $P(Y,Z)>0$.
2. $P(X|Y,Z) = f(X,Z)$ *for some functions* f.
3. $P(X,Y|Z) = P(X|Z)P(Y|Z)$ *if* $P(Z)>0$.
4. $P(X,Y|Z) = f(X,Z)g(Y,Z)$ *for some functions* f *and* g.
5. $P(X,Y,Z) = P(X|Z)P(Y|Z)P(Z)$ *if* $P(Z)>0$.
6. $P(X,Y,Z) = P(X,Z)P(Y,Z)/P(Z)$ *if* $P(Z)>0$.
7. $P(X,Y,Z) = f(X,Z)g(Y,Z)$ **for some functions** f **and** g**.**

Exercise: Prove the theorem.

Exam3. Problem 1.







Day: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Weather: S S R C C R R S S S R C C R S S C

Given above sequence of weather events, estimate the transition probability A:

$$A_{x_{t-1},x_t} = P(x_t \mid x_{t-1})$$

Homework: Compute conditional transition probabilities (CTP).

$P(x_t = s \mid x_{t-1} = s) = 4/7$
$P(x_t = c \mid x_{t-1} = c) = 2/5$
$P(x_t = r \mid x_{t-1} = c) = 2/5$

Independence

- Two variables are *independent* in a joint distribution if:

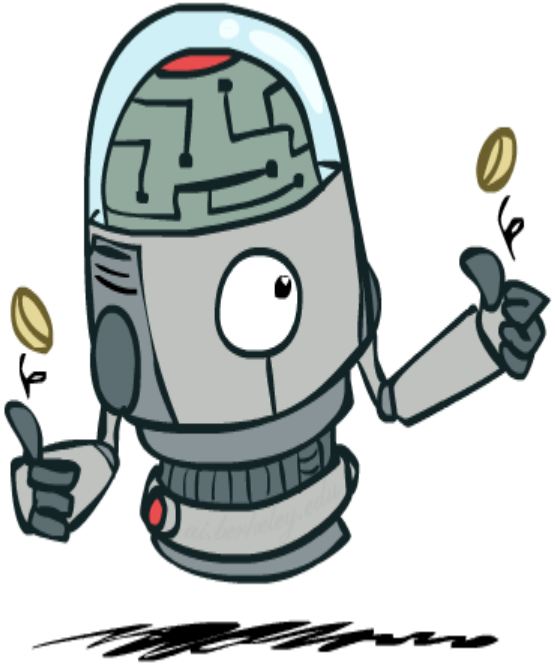
$$P(X,Y) = P(X)P(Y)$$

$$X \perp\!\!\!\perp Y$$

$$\forall x, y \, P(x,y) = P(x)P(y)$$

  - Says the joint distribution *factors* into a product of two simple ones
  - Usually variables aren't independent!

- Can use independence as a *modeling assumption*
  - Independence can be a simplifying assumption
  - *Empirical* joint distributions: at best "close" to independent



Conditional Independence

- Unconditional (absolute) independence very rare (why?)
- *Conditional independence* is our most basic and robust form of knowledge about uncertain environments.

$$X \perp\!\!\!\perp Y | Z$$

- X is conditionally independent of Y given Z

if and only if:

$$\forall x, y, z : P(x, y|z) = P(x|z)P(y|z)$$

or, equivalently, if and only if

$$\forall x, y, z : P(x|z, y) = P(x|z)$$

Probability Recap

- Conditional probability $$P(x|y) = \frac{P(x,y)}{P(y)}$$
- Product rule $$P(x,y) = P(x|y)P(y)$$
- Chain rule $$\begin{aligned} P(X_1, X_2, \ldots X_n) &= P(X_1)P(X_2|X_1)P(X_3|X_1, X_2)\ldots \\ &= \prod_{i=1}^{n} P(X_i|X_1, \ldots, X_{i-1}) \end{aligned}$$
- X, Y independent if and only if: $$\forall x, y : P(x,y) = P(x)P(y)$$
- X and Y are conditionally independent given Z if and only if:

$$\forall x, y, z : P(x,y|z) = P(x|z)P(y|z) \qquad X \perp\!\!\!\perp Y | Z$$

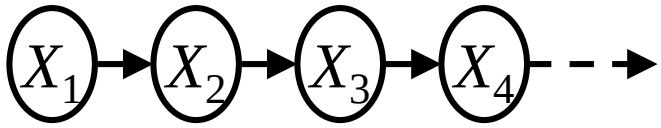
Reasoning over Time or Space

- Often, we want to reason about a sequence of observations
  - Speech recognition
  - Robot localization
  - User attention
  - Medical monitoring

- Need to introduce time (or space) into our models

Markov Models

– Value of X at a given time is called the state

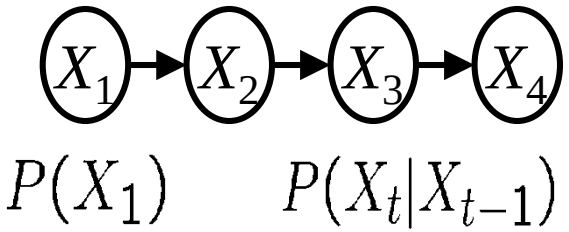


$$P(X_1) \qquad P(X_t|X_{t-1})$$

– Parameters: called transition probabilities or dynamics, specify how the state evolves over time (also, initial state probabilities)

– Stationarity assumption: transition probabilities the same at all times

Joint Distribution of a Markov Model



- Joint distribution:

$$P(X_1, X_2, X_3, X_4) = P(X_1)P(X_2|X_1)P(X_3|X_2)P(X_4|X_3)$$

- More generally:

$$P(X_1, X_2, \ldots, X_T) = P(X_1)P(X_2|X_1)P(X_3|X_2)\ldots P(X_T|X_{T-1})$$

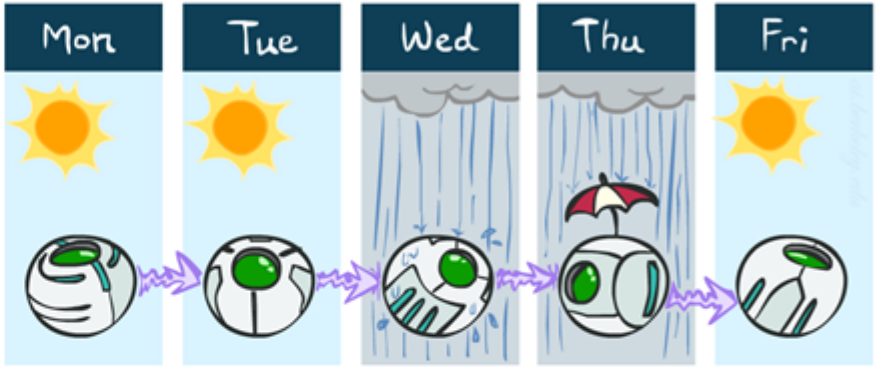
$$= P(X_1)\prod_{t=2}^{T} P(X_t|X_{t-1})$$

- Questions to be resolved:
  - Does this indeed define a joint distribution?
  - Can every joint distribution be factored this way, or are we making some assumptions about the joint distribution by using this factorization?

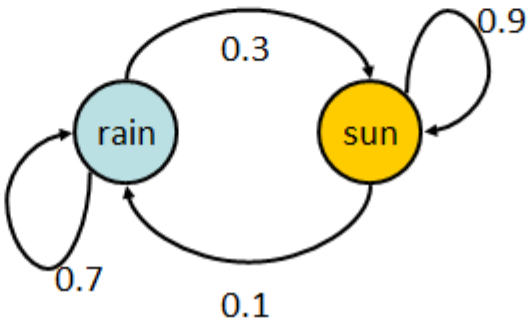
Example Markov Chain: Weather

- States: X = {rain, sun}
- Initial distribution: 1.0 sun
- CPT $P(X_t \mid X_{t-1})$:

X_{t-1}	X_t	$P(X_t \mid X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7



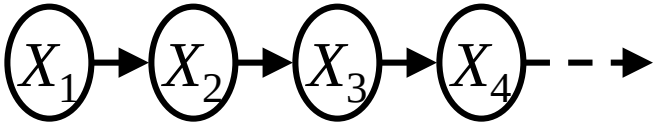
Two new ways of representing the same CPT



$$P(x_t|x_{t-1}) = A_{x_{t-1},x_t}$$

$$A = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix}$$

Forward Propagation



$P(x_1) = \text{known}$

Marginal distribution equals to sum of joint distribution

$$P(x_t) = \sum_{x_{t-1}} P(x_{t-1}, x_t)$$

$$= \sum_{x_{t-1}} P(x_t \mid x_{t-1}) P(x_{t-1})$$

$$\pi^{x_t} = \sum_{x_{t-1}} A_{x_{t-1}, x_t} \pi^{x_{t-1}}$$

$$\pi_t^x = \sum_{x'} A_{x',x} \pi_{t-1}^{x'}$$

$$\pi_t = \pi_{t-1} A$$

Let $\pi_t = (\pi^S, \pi^R, \pi^C)$

represent today's weather distribution.

Tommorrow's weather distribution π_{t+1}

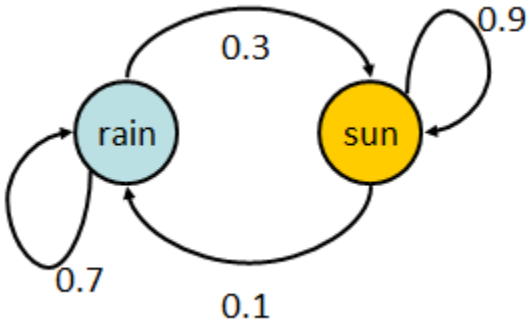
$$\pi_{t+1} = \pi_t A$$

For example,

$$\pi_{t+1}^S = \pi_t^S A_{SS} + \pi_t^R A_{RS} + \pi_t^C A_{CS}$$

Markov Chain Propagation

Weather problem



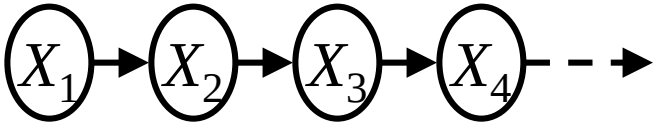
- Initial distribution:
- $P(X_1 = \text{sun}) = 1.0$

- Probability distribution after one step:

$$P(X_2 = \text{sun}) = P(X_2 = \text{sun}|X_1 = \text{sun})P(X_1 = \text{sun}) + P(X_2 = \text{sun}|X_1 = \text{rain})P(X_1 = \text{rain})$$

$$= 0.9 \cdot 1.0 + 0.3 \cdot 0.0 = 0.9$$

Forward Propagation



Let $\pi_t = (\pi^S, \pi^R, \pi^C)$ represent today's weather distribution.

Tommorrow's weather distribution π_{t+1} is $\pi_{t+1} = \pi_t A$

We can write $\pi_{t+1} = \pi_t A = (\pi_{t-1} A)A = \pi_{t-1} A^2$

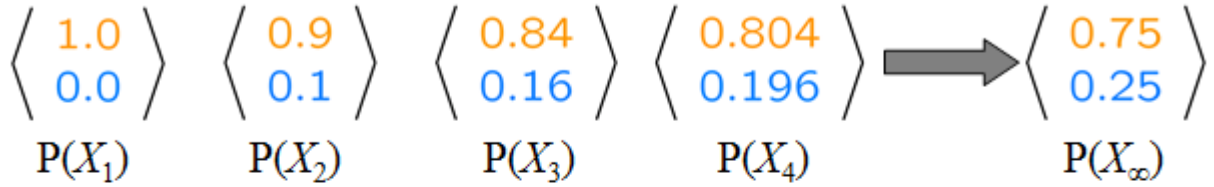
$\pi_{t+k} = \pi_t A^k$ for the next k-th day weather distribution

When k goes large, the weather approaches stationary distribution

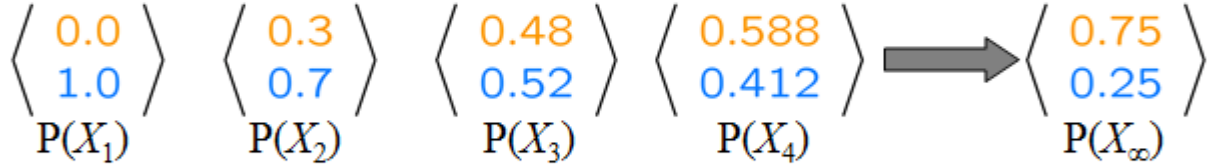
$$\pi_\infty A = \pi_{\infty+1} = \pi_\infty$$

Run of Propagation towards infinity

- From initial observation of sunny



- From initial observation of rainny



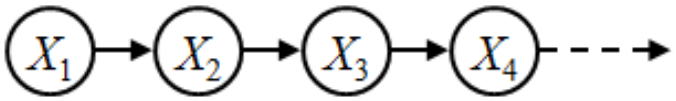
- From yet another initial distribution $P(X_1)$



Compute Stationary Distribution

- Question: What's P(X) at time t = infinity?

$$\pi_\infty A = \pi_{\infty+1} = \pi_\infty$$



$$P_\infty(sun) = P(sun|sun)P_\infty(sun) + P(sun|rain)P_\infty(rain)$$
$$P_\infty(rain) = P(rain|sun)P_\infty(sun) + P(rain|rain)P_\infty(rain)$$

$$P_\infty(sun) = 0.9P_\infty(sun) + 0.3P_\infty(rain)$$
$$P_\infty(rain) = 0.1P_\infty(sun) + 0.7P_\infty(rain)$$

$$P_\infty(sun) = 3P_\infty(rain)$$
$$P_\infty(rain) = 1/3P_\infty(sun)$$

Also: $P_\infty(sun) + P_\infty(rain) = 1$

$$P_\infty(sun) = 3/4$$
$$P_\infty(rain) = 1/4$$

X_{t-1}	X_t	$P(X_t \mid X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7

Stationary Distribution 稳态分布

- For most chains:
  - Influence of the initial distribution gets less and less over time.
  - The distribution we end up in is independent of the initial distribution

- Stationary distribution:
  - The distribution we end up with is called the stationary distribution P_∞ of the chain
  - It satisfies

$$P_\infty(X) = P_{\infty+1}(X) = \sum_x P(X|x)P_\infty(x)$$

Ingredients of a Markov Model

- States:

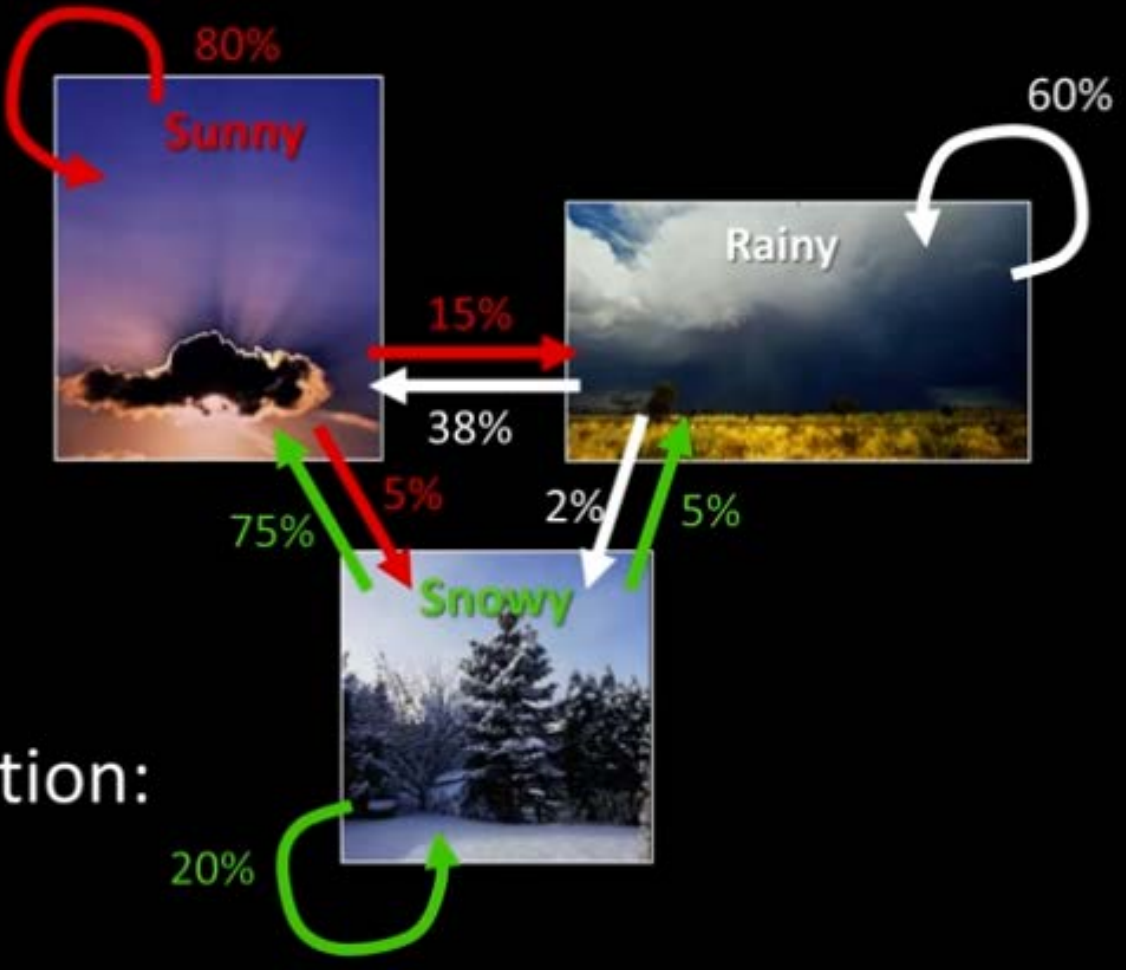
$$\{S_{sunny}, S_{rainy}, S_{snowy}\}$$

- State transition
- probabilities:

$$A = \begin{pmatrix} .8 & .15 & .05 \\ .38 & .6 & .02 \\ .75 & .05 & .2 \end{pmatrix}$$

- Initial state distribution:

$$\pi = (.7 \quad .25 \quad .05)$$



Probability of a Time Series

- Given:



- What is the probability of this series?

$$P(S_{sunny}) \cdot P(S_{rainy} \mid S_{sunny}) \cdot P(S_{rainy} \mid S_{rainy}) \cdot P(S_{rainy} \mid S_{rainy})$$

$$\cdot P(S_{snowy} \mid S_{rainy}) \cdot P(S_{snowy} \mid S_{snowy})$$

$$= 0.7 \cdot 0.15 \cdot 0.6 \cdot 0.6 \cdot 0.02 \cdot 0.2 = 0.0001512$$

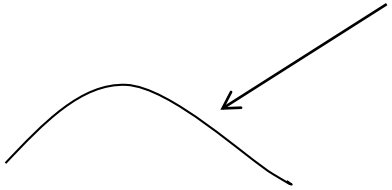
$$A = \begin{pmatrix} .8 & .15 & .05 \\ .38 & .6 & .02 \\ .75 & .05 & .2 \end{pmatrix} \qquad \pi = (.7 \quad .25 \quad .05)$$

Homework:

What is the most probable 4 day weather sequence? Assume weather={**S**unny, C**l**oudy , **R**aining} with transition probability matrix:

$$A = \begin{pmatrix} .8 & .15 & .05 \\ .38 & .6 & .02 \\ .75 & .05 & .2 \end{pmatrix}$$

4-day sequence can start any day



W1 W2 W3 W4 W5 W6 W7 W8 W9 W10 W11 W12 … … …

Hint: today's distribution is the stationary distribution

If today is sunny, will tommorrow be sunny?
How many continuous sunny days we should expect?



.....



1 more day, A_{kk},
2 more days, A_{kk}^2,
3 more days, A_{kk}^3,

k=Sunny

$$A = \begin{pmatrix} .8 & .15 & .05 \\ .38 & .6 & .02 \\ .75 & .05 & .2 \end{pmatrix}$$

Average = $$\frac{1A_{kk} + 2A_{kk}^2 + 3A_{kk}^3 + \cdots}{A_{kk} + A_{kk}^2 + A_{kk}^3 + \cdots} = \frac{1 + 2A_{kk} + 3A_{kk}^2 + \cdots}{1 + A_{kk} + A_{kk}^2 + \cdots} = \frac{1/(1 - A_{kk})^2}{1/(1 - A_{kk})} = \frac{1}{1 - A_{kk}}$$

Thus we expect $\frac{1}{1 - A_{kk}}$ more sunny days

Homework

If today is rainny, will tommorrow be sunny? Or the day after tommorrow be sunny ?

If today is rainny, how many days I need to wait to get a sunny day?

$$A = \begin{pmatrix} .8 & .15 & .05 \\ .38 & .6 & .02 \\ .75 & .05 & .2 \end{pmatrix}$$

Some possible sequences:

























Tuesday, March 24, Homework for probabilistic-reasoning is assigned.
Thursday, March 26, No class
Tuesday March 31, A brief review/highlights of class lectures

Bayesian Network

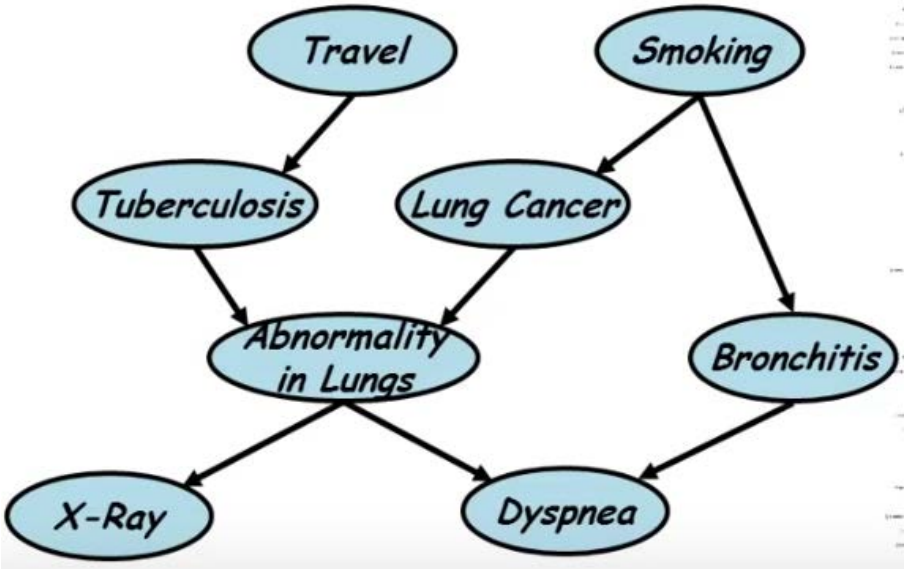
Many Applications

- Medical diagnosis
- Fault diagnosis
- Natural language processing
- Traffic analysis
- Social network models
- Message decoding
- Computer vision
  - Image segmentation
  - 3D reconstruction
  - Holistic scene analysis
- Speech recognition
- Robot localization & mapping
- Genomics

A Bayesian Network is a directed acyclic graph (DAG), where each node is a random variable



predisposing factors
symptoms
test results
diseases
treatment outcomes



From
Full probability distribution to
Bayesian Networks

Joint probability distribution

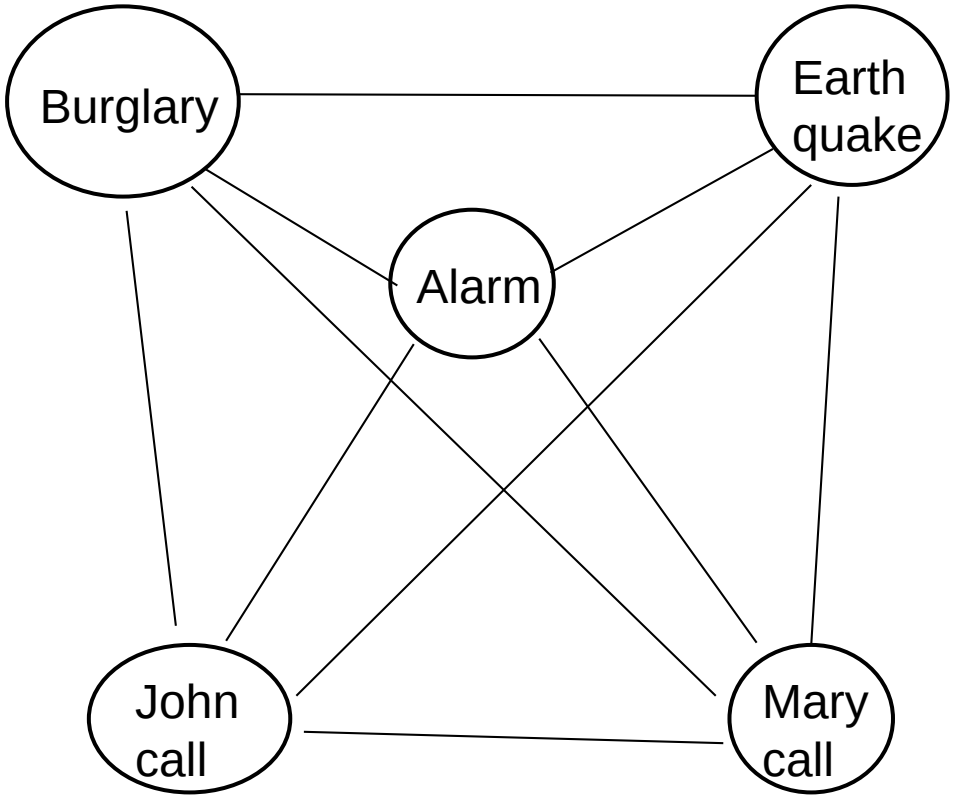
$P(B, E, A, J, M)$

B	E	A	J	M	Prob	B	E	A	J	M	Prob
y	y	y	y	y	.00001	n	y	y	y	y	.0002
y	y	y	y	n	.000025	n	y	y	y	n	.0004
y	y	y	n	y	.000025	n	y	y	n	y	.0004
y	y	y	n	n	.00000	n	y	y	n	n	.0002
y	y	n	y	y	.00001	n	y	n	y	y	.0002
y	y	n	y	n	.000015	n	y	n	y	n	.0002
y	y	n	n	y	.000015	n	y	n	n	y	.0002
y	y	n	n	n	.0000	n	y	n	n	n	.0002
y	n	y	y	y	.00001	n	n	y	y	y	.0001
y	n	y	y	n	.000025	n	n	y	y	n	.0002
y	n	y	n	y	.000025	n	n	y	n	y	.0002
y	n	y	n	n	.0000	n	n	y	n	n	.0001
y	n	n	y	y	.00001	n	n	n	y	y	.0001
y	n	n	y	n	.00001	n	n	n	y	n	.0001
y	n	n	n	y	.00001	n	n	n	n	y	.0001
y	n	n	n	n	.00000	n	n	n	n	n	.996

Assume you live in Los Angeles. You have a house and has installed an alarm system. You have two neighbors, John living in one house and Mary living in another house. They are retired and mostly stay at home. Earth quake occurs in L.A. Burglary also happens too. When alarm is turned on, your neighbors John, or Mary often calls you while you are at work.

These 5 random variables form a joint probability distribution.

Prob(B,E,A,J,M)



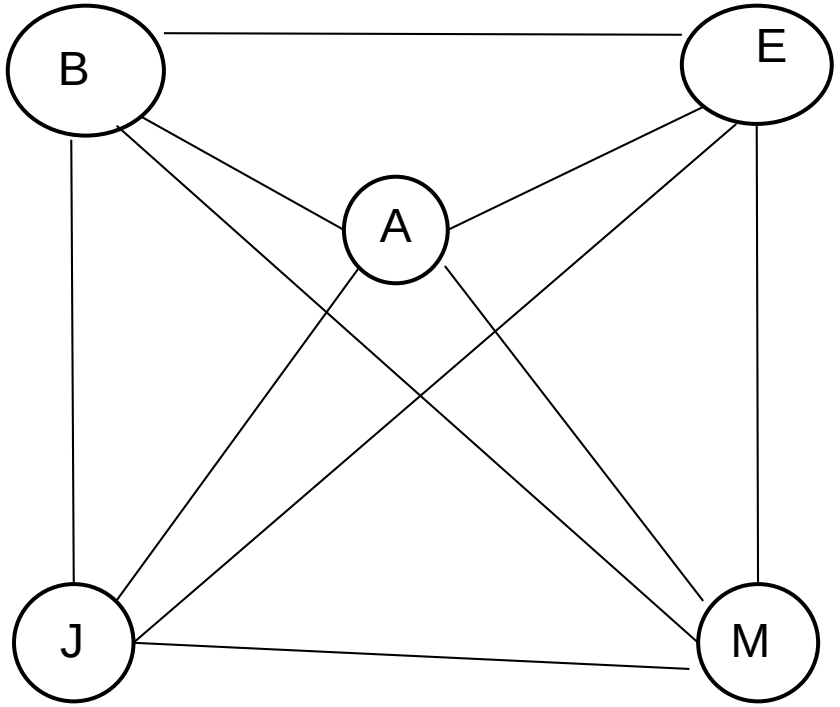
B	E	A	J	M	Prob
y	y	y	y	y	.00001
y	y	y	y	n	.000025
y	y	y	n	y	.000025
y	y	y	n	n	.00000
y	y	n	y	y	.00001
y	y	n	y	n	.000015
y	y	n	n	y	.000015
y	y	n	n	n	.0000
y	n	y	y	y	.00001
y	n	y	y	n	.000025
y	n	y	n	y	.000025
y	n	y	n	n	.0000
y	n	n	y	y	.00001
y	n	n	y	n	.00001
y	n	n	n	y	.00001
y	n	n	n	n	.00000
n	y	y	y	y	.0002
n	y	y	y	n	.0004
n	y	y	n	y	.0004
n	y	y	n	n	.0002
n	y	n	y	y	.0002
n	y	n	y	n	.0002
n	y	n	n	y	.0002
n	y	n	n	n	.0002
n	n	y	y	y	.0001
n	n	y	y	n	.0002
n	n	y	n	y	.0002
n	n	y	n	n	.0001
n	n	n	y	y	.0001
n	n	n	y	n	.0001
n	n	n	n	y	.0001
n	n	n	n	n	.996

Full probability/distribution

Five random variables form a joint probability distribution.

If we don't know the internal structure, we need $2^5 - 1 = 31$ numbers to specify the distribution.

Prob(B,E,A,J,M)



B	E	A	J	M	Prob
y	y	y	y	y	.00001
y	y	y	y	n	.000025
y	y	y	n	y	.000025
y	y	y	n	n	.00000
y	y	n	y	y	.00001
y	y	n	y	n	.000015
y	y	n	n	y	.000015
y	y	n	n	n	.0000
y	n	y	y	y	.00001
y	n	y	y	n	.000025
y	n	y	n	y	.000025
y	n	y	n	n	.0000
y	n	n	y	y	.00001
y	n	n	y	n	.00001
y	n	n	n	y	.00001
y	n	n	n	n	.00000
n	y	y	y	y	.0002
n	y	y	y	n	.0004
n	y	y	n	y	.0004
n	y	y	n	n	.0002
n	y	n	y	y	.0002
n	y	n	y	n	.0002
n	y	n	n	y	.0002
n	y	n	n	n	.0002
n	n	y	y	y	.0001
n	n	y	y	n	.0002
n	n	y	n	y	.0002
n	n	y	n	n	.0001
n	n	n	y	y	.0001
n	n	n	y	n	.0001
n	n	n	n	y	.0001
n	n	n	n	n	.996

Inference on full distribution is down easily

- What is the probability of burglary given that Mary called, $P(B{=}y|M{=}y)$?
- Compute *marginal probability*:

$$P(B, M) = \sum_{E,A,J} P(B, E, A, J, M)$$

B	M	Prob
y	y	.000115
y	n	.000075
n	y	.00015
n	n	.99966

- Compute answer (reasoning by conditioning):

$$\begin{aligned} P(B{=}y|M{=}y) &= \frac{P(B{=}y, M{=}y)}{P(M{=}y)} \\ &= \frac{.000115}{.000115 + 000075} = 0.61 \end{aligned}$$

- Probability theory well-established and well-understood.
- In theory, can perform arbitrary inference among the variables given a joint probability. This is because the joint probability contains information of all aspects of the relationships among the variables.
  - Diagnostic inference:
    - From effects to causes.
    - Example: $P(B{=}y|M{=}y)$
  - Predictive inference:
    - From causes to effects.
    - Example: $P(M{=}y|B{=}y)$
  - Combining evidence:

$$P(B{=}y|J{=}y, M{=}y, E{=}n)$$

Using conditional independence,
reduce full probability distribution to
Bayesian Networks

Markov Model: conditional independence







Day: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 …

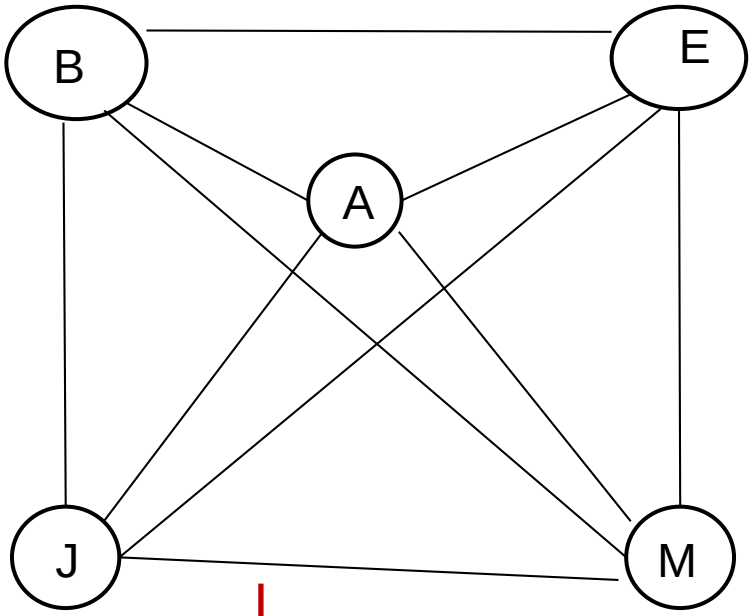
Weather: S S R C C R R S S S R C C R S …

Markov assumption:
tommorrow's weather
depends only on today's weather,
independent of yesterday's weather, day before yesterday's weather, etc

$$P(X_{t+1} \mid X_t, X_{t-1}, X_{t-2}, \ldots) = P(X_{t+1} \mid X_t, \cancel{X_{t-1}}, \cancel{X_{t-2}}, \ldots)$$

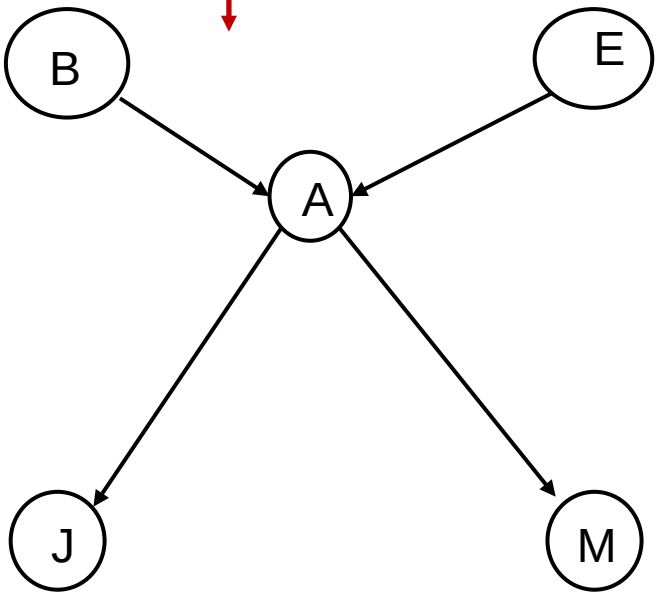
Prob(B,E,A,J,M)

Full model



Internal structure, causal relations.

Simplified model



B	E	A	J	M	Prob
y	y	y	y	y	.00001
y	y	y	y	n	.000025
y	y	y	n	y	.000025
y	y	y	n	n	.00000
y	y	n	y	y	.00001
y	y	n	y	n	.000015
y	y	n	n	y	.000015
y	y	n	n	n	.0000
y	n	y	y	y	.00001
y	n	y	y	n	.000025
y	n	y	n	y	.000025
y	n	y	n	n	.0000
y	n	n	y	y	.00001
y	n	n	y	n	.00001
y	n	n	n	y	.00001
y	n	n	n	n	.00000
n	y	y	y	y	.0002
n	y	y	y	n	.0004
n	y	y	n	y	.0004
n	y	y	n	n	.0002
n	y	n	y	y	.0002
n	y	n	y	n	.0002
n	y	n	n	y	.0002
n	y	n	n	n	.0002
n	n	y	y	y	.0001
n	n	y	y	n	.0002
n	n	y	n	y	.0002
n	n	y	n	n	.0001
n	n	n	y	y	.0001
n	n	n	y	n	.0001
n	n	n	n	y	.0001
n	n	n	n	n	.996

Full probability model needs 31 parameters.

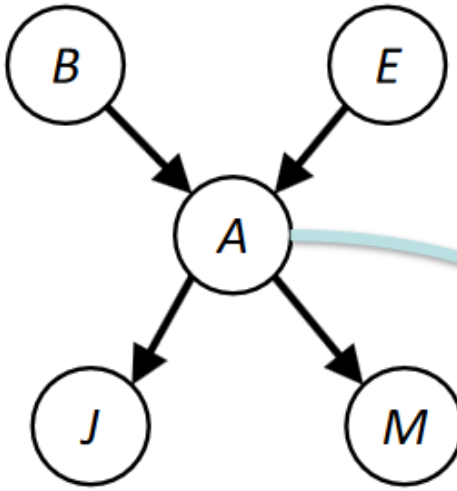
Knowning the conditional indep. structure, we need

$$(2-1) + (2-1) + 2*(2-1) + 2*(2-1) + 2*2*(2-1) = 1+1+2+2+4=10$$

parameters to specify the distribution.

Greatly simplies the model and inference

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

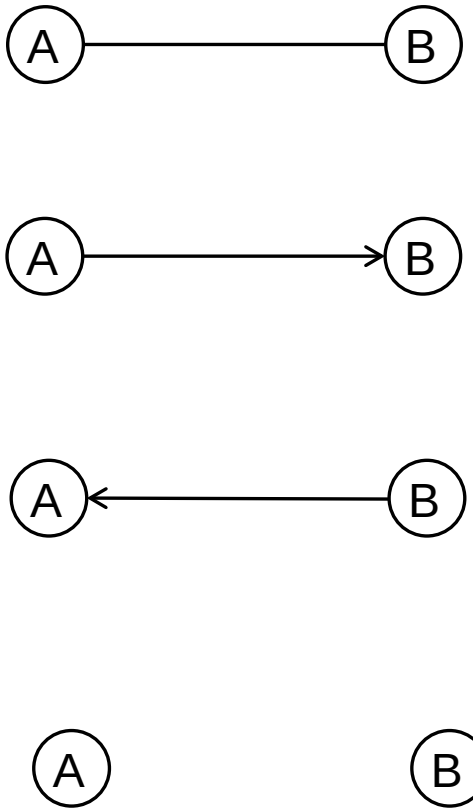


B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01

For two random variables, network structure is flexible



Equavalent conditions for conditional independence

Proposition (1.1)

Variables X and Y are conditionally independent given Z if and only if one of the following conditions is met:

1. $P(X|Y,Z) = P(X|Z)$ *if* $P(Y,Z)>0$. This is the starting assumption
2. $P(X|Y,Z) = f(X,Z)$ *for some functions* f.
3. $P(X,Y|Z) = P(X|Z)P(Y|Z)$ *if* $P(Z)>0$. Another useful equivalent form
4. $P(X,Y|Z) = f(X,Z)g(Y,Z)$ *for some functions* f *and* g.
5. $P(X,Y,Z) = P(X|Z)P(Y|Z)P(Z)$ *if* $P(Z)>0$.
6. $P(X,Y,Z) = P(X,Z)P(Y,Z)/P(Z)$ *if* $P(Z)>0$.
7. $P(X,Y,Z) = f(X,Z)g(Y,Z)$ **for some functions** f **and** g**.**

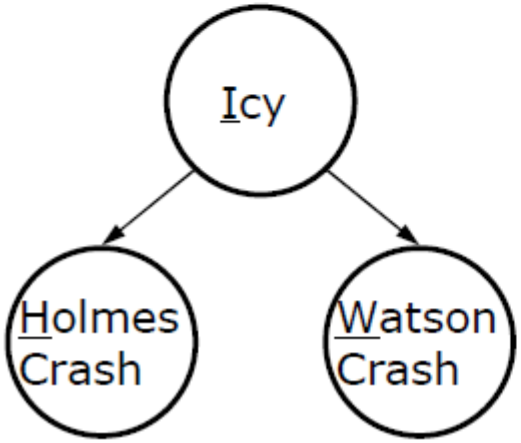
Exercise: Prove the theorem.

Well known and most inspiring Bayesian Network Examples

Icy Roads

Inspector Smith is waiting for Holmes and Watson, who are driving (separately) to meet him. It is winter. His secretary tells him that Watson has had an accident. He says, "It must be that the roads are icy. I bet that Holmes will have an accident too. I should go to lunch." But, his secretary says, "No, the roads are not icy, look at the window." So, he says, "I guess I better wait for Holmes."

"Causal" Component

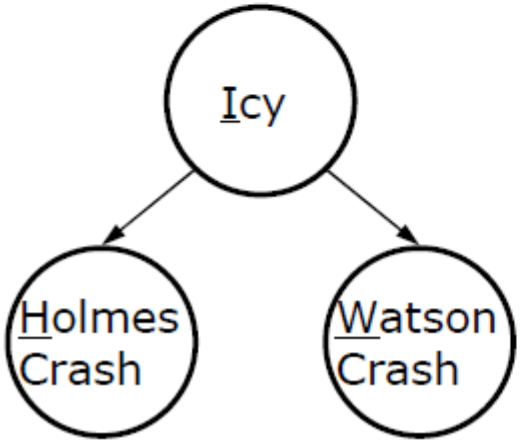


Probability distribution: $P(H,W,I) = P(H|I)\ P(W|I)\ P(I)$

Icy Roads

Inspector Smith is waiting for Holmes and Watson, who are driving (separately) to meet him. It is winter. His secretary tells him that Watson has had an accident. He says, "It must be that the roads are icy. I bet that Holmes will have an accident too. I should go to lunch." But, his secretary says, "No, the roads are not icy, look at the window." So, he says, "I guess I better wait for Holmes."

"Causal" Component



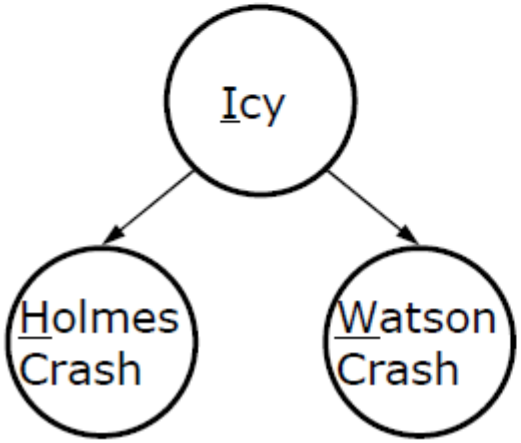
At this time, Smith thinks H and W are dependent.
Reasoning: Watson crashed --> Icy more likely --> Holmes more likely to crash.

Here Icy variable is not known at beginning, and Holmes and Watson are dependent.

Icy Roads

Inspector Smith is waiting for Holmes and Watson, who are driving (separately) to meet him. It is winter. His secretary tells him that Watson has had an accident. He says, "It must be that the roads are icy. I bet that Holmes will have an accident too. I should go to lunch." But, his secretary says, "No, the roads are not icy, look at the window." So, he says, "I guess I better wait for Holmes."

"Causal" Component



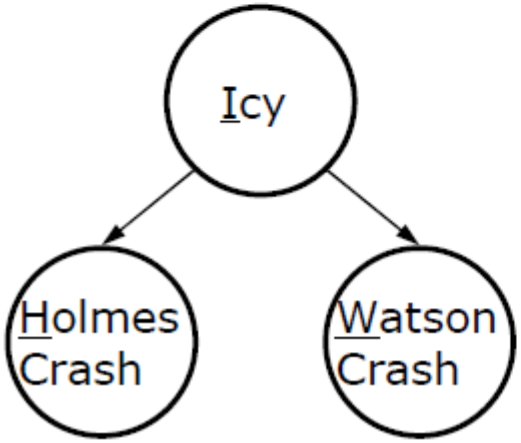
At this time, Smith learned the fact that road is not icy (i.e., Icy variable is set to 0), and he asserts that H and W are independent.

When Icy is set (either 0 or 1), Watson and Holmes are independent.

Icy Roads

Inspector Smith is waiting for Holmes and Watson, who are driving (separately) to meet him. It is winter. His secretary tells him that Watson has had an accident. He says, "It must be that the roads are icy. I bet that Holmes will have an accident too. I should go to lunch." But, his secretary says, "No, the roads are not icy, look at the window." So, he says, "I guess I better wait for Holmes."

"Causal" Component

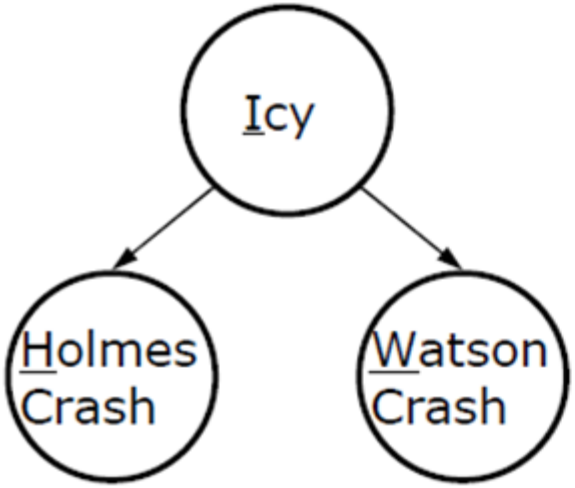


Conclusion:

H and W are dependent when I is not known.

H and W are independent when I is known.

Conditional independence



Conclusion:

H and W are dependent when I is not known.

H and W are independent when I is known.

Why???

$$P(H,W,I) = P(H|I)\,P(W|I)\,P(I)$$

This decomposition follows from the diagram

Suppose $I = I_0$ is known,

$$P(H,W|I_0) = P(H|I_0)\,P(W|I_0)\,P(I_0)/P(I_0)$$
$$= P(H|I_0)\,P(W|I_0)$$

Thus, H and W become **conditionally-independent**.

This is called d-separation of H, W by the fact Icy is fixed to a value.

Suppose I is not known.

Knowing W=1 will increase Prob(Icy=1), which, in turn, will increase Prob(H=1). Thus W and H are dependent.

House Lawn Wet. Explained Away!

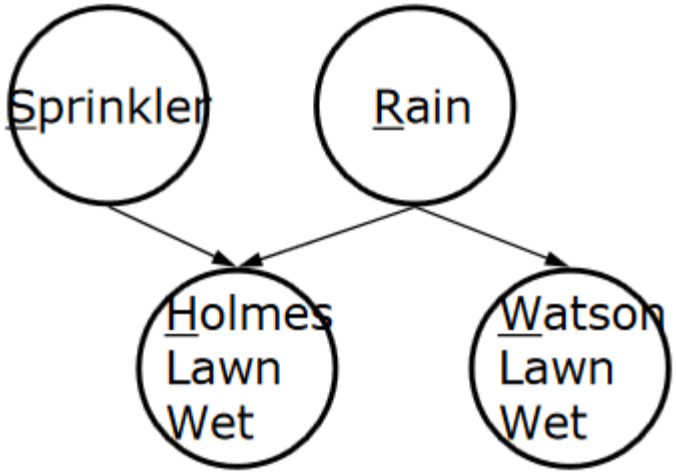
Holmes and Watson in LA

One morning Holmes

Holmes and Watson have moved to LA. wakes up to find his lawn wet. He wonders if it has rained or if he left his sprinkler on. He looks at his neighbor Watson's lawn and he sees it is wet too. So, he concludes it must have rained.

At this time, Prob(Sprinkler) and Prob(Rain) both increased.

At this time, Prob(Rain) increased and Prob(Sprinkler) decreased.



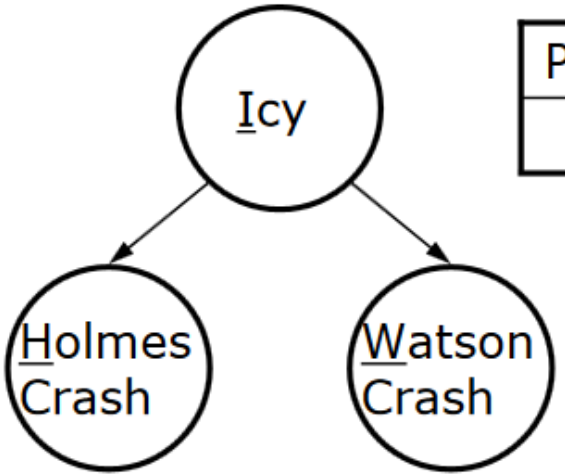
Holmes' worry that he forgot to turn off the sprinkler was explained away !

Probability distribution: $P(S,R,H,W) = P(H|S,R)\ P(W|R)\ P(S)\ P(R)$

Icy Roads with Numbers

t= true

f= false

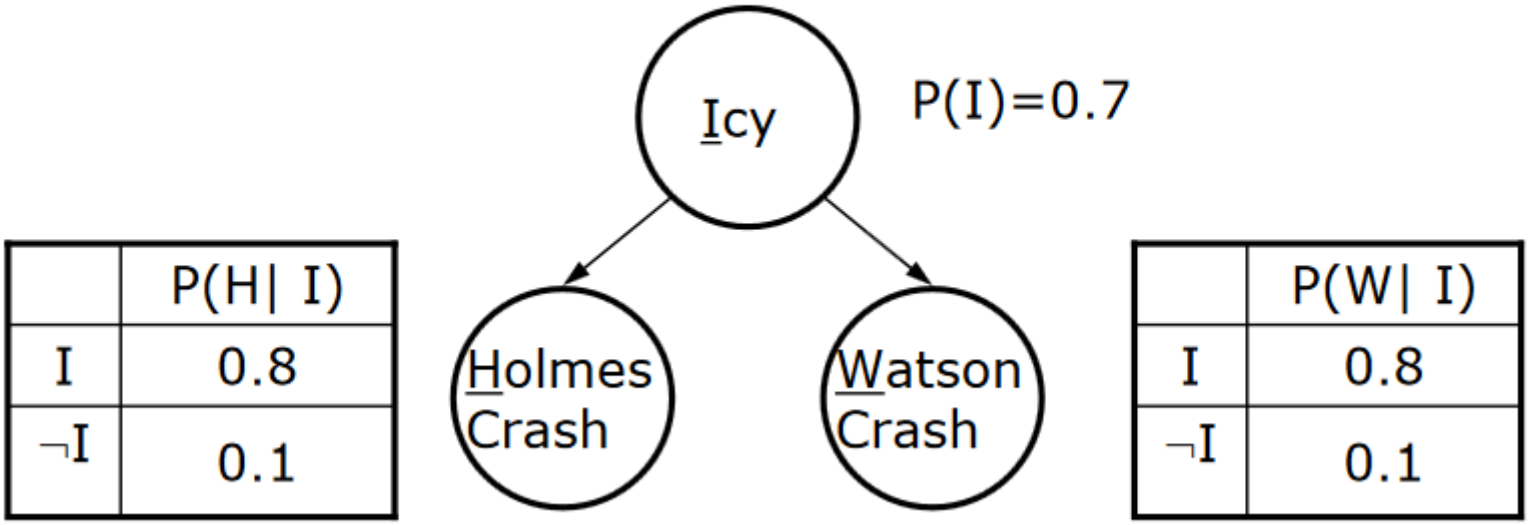


P(I=t)	P(I=f)
0.7	0.3

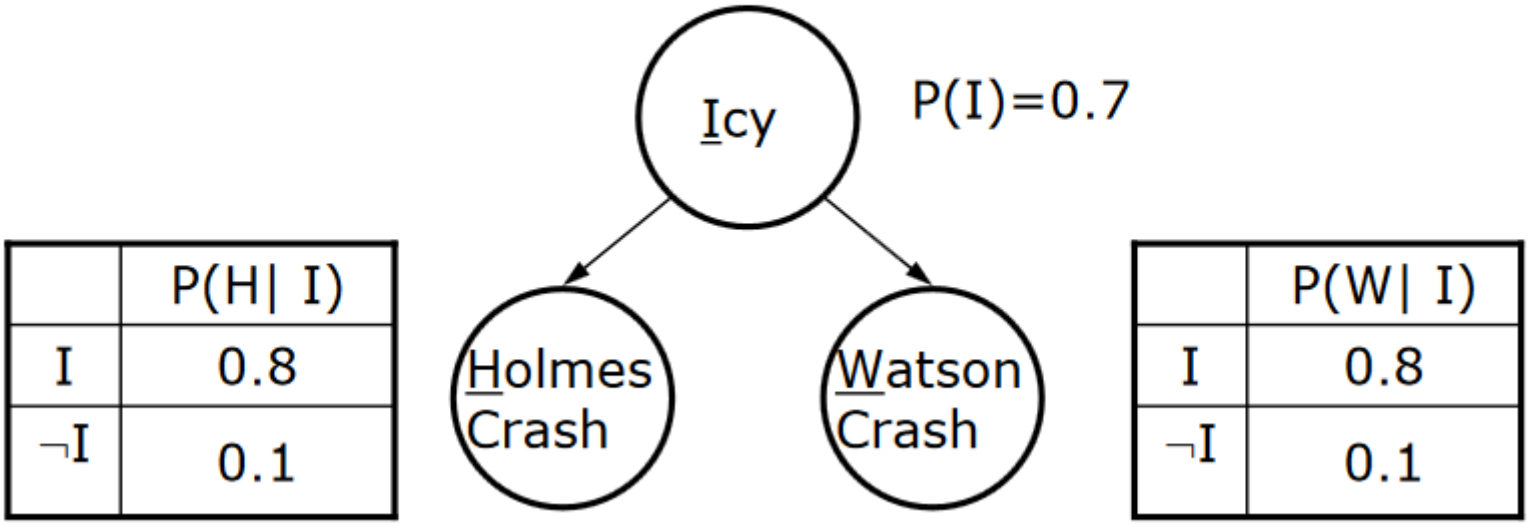
	P(H=t | I)	P(H=f | I)
I=t	0.8	0.2
I=f	0.1	0.9

	P(W=t | I)	P(W=f | I)
I=t	0.8	0.2
I=f	0.1	0.9

Numerical Example: Shorthand



Numerical Example: Shorthand



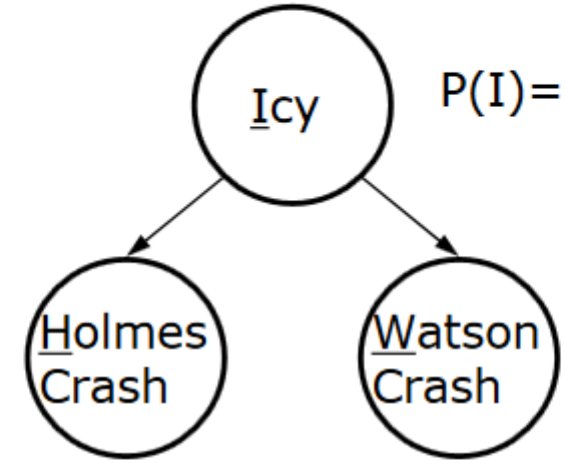
	P(H| I)
I	0.8
¬I	0.1

	P(W| I)
I	0.8
¬I	0.1

We will show

1. Without knowning anything, Holmes crash prob = P(H) = P(W) =0.59
2. If we know Watson crash, then prob of Icy increased from 0.7 to 0.95, and prob of Holmes crash increased from 0.59 to 0.765
3. This also shows that when Icy is unknown, H,W are related.
4. Of course, when Icy is set/known, H,W occur according to the given known probability --- conditional independent.

Probability that Watson Crashes

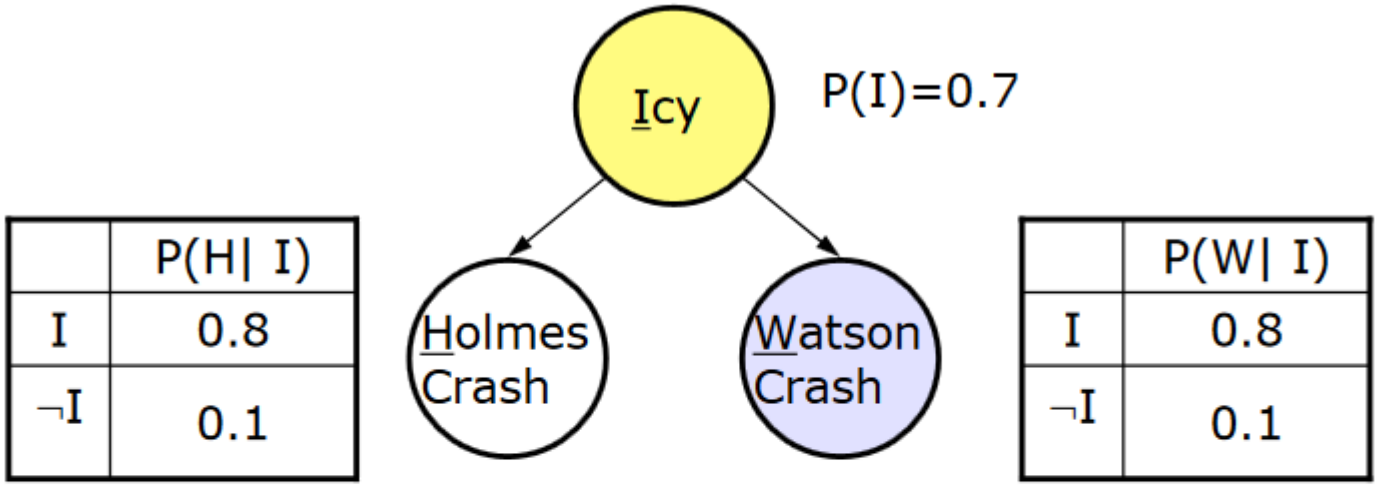


	P(H| I)
I	0.8
¬I	0.1

	P(W| I)
I	0.8
¬I	0.1

$$P(W) = P(W|I)\,P(I) + P(W|\neg I)\,P(\neg I)$$
$$= 0.8 \cdot 0.7 + 0.1 \cdot 0.3$$
$$= 0.56 + 0.03$$
$$= 0.59$$

Probability of Icy given Watson



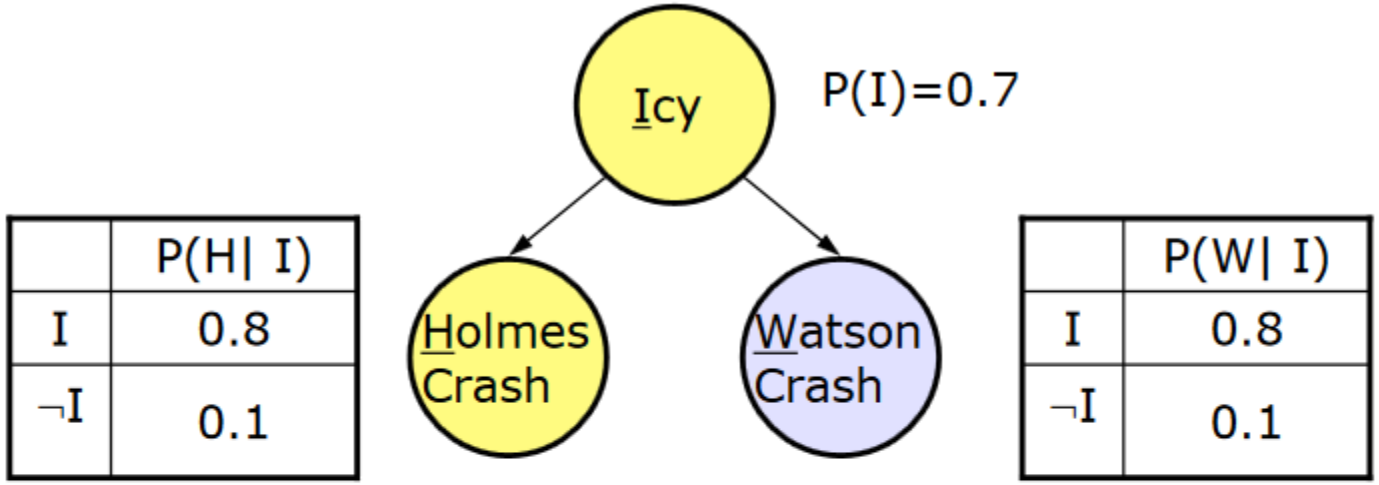
	P(H| I)
I	0.8
¬I	0.1

	P(W| I)
I	0.8
¬I	0.1

$$P(I \mid W) = P(W \mid I)\, P(I) / P(W)$$
$$= 0.8 \cdot 0.7 / 0.59$$
$$= 0.95$$

We started with P(I) = 0.7; knowing that Watson crashed raised the probability to 0.95

Probability of Holmes given Watson

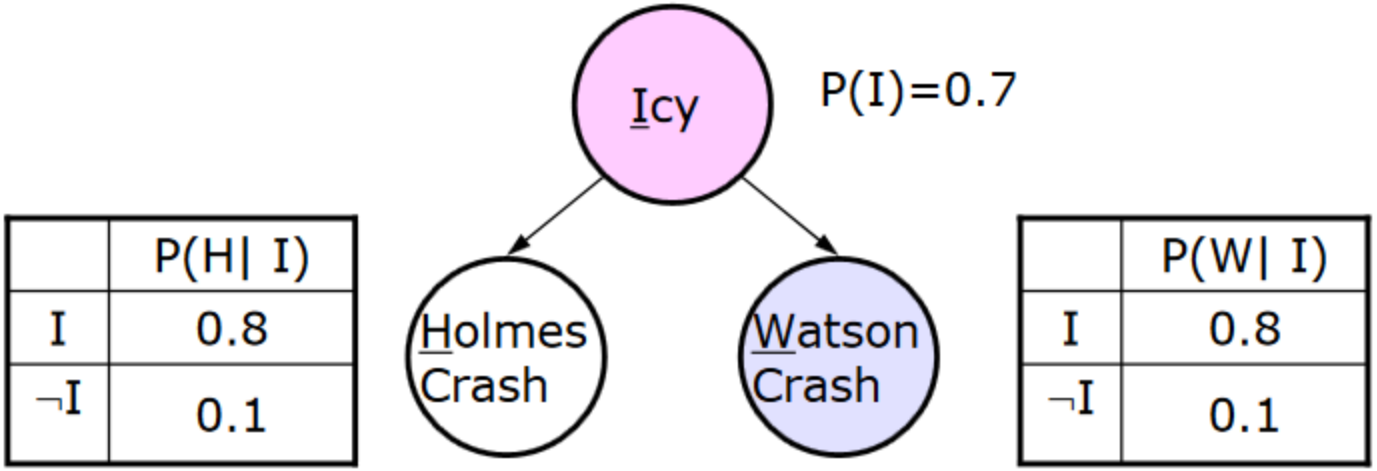


	P(H| I)
I	0.8
¬I	0.1

	P(W| I)
I	0.8
¬I	0.1

$$
\begin{aligned}
P(H|W) &= P(H|W,I)P(I|W) + P(H|W,\neg I)\,P(\neg I|\,W) \\
&= P(H|I)P(I|W) + P(H|\neg I)\,P(\neg I|\,W) \\
&= 0.8 \cdot 0.95 + 0.1 \cdot 0.05 \\
&= 0.765
\end{aligned}
$$

We started with P(H) = 0.59; knowing that Watson crashed raised the probability to 0.765



$P(H|W, \neg I) = P(H|\neg I) = 0.1$

H and W are d-separated given I, so H and W are conditionally independent given I

Use Bayes theorem, we can obtain the same conclusion **without assuming d-separation**.

House Lawn Wet. Explained Away!

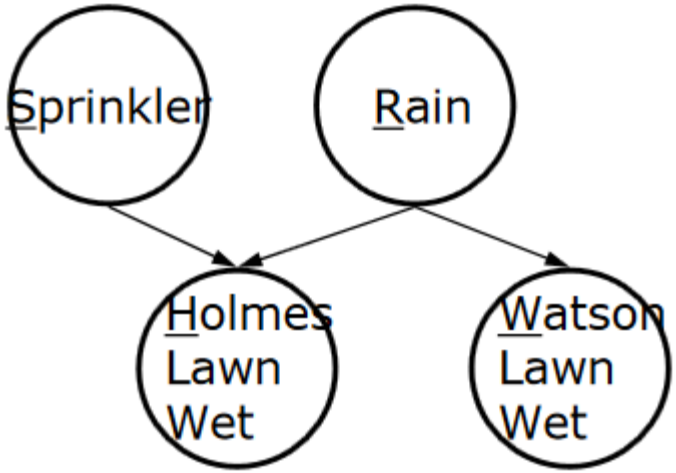
Holmes and Watson in LA

One morning Holms wakes up to find his lawn wet. He wonders if it has rained or if he left his sprinkler on. He looks at his neighbor Watson's lawn and he sees it is wet too. So, he concludes it must have rained.

Holmes and Watson have moved to LA.

At this time, Prob(Sprinkler) and Prob(Rain) both increased.

At this time, Prob(Rain) increased and Prob(Sprinkler) decreased.



The worry that Homs forgot to turn off the sprinkler was explained away !

Probability distribution: $P(S,R,H,W) = P(H|S,R)\ P(W|R)\ P(S)\ P(R)$

House Lawn Wet. Explained Away!

Homework:

In the Watson and Holmes visit LA network, use the following conditional probability tables.

P(R) = 0.2

P(S) = 0.1

	P(W| R)
R	1.0
¬R	0.2

	P(H| R,S)
R,S	1.0
R,¬S	1.0
¬ R,S	0.9
¬ R,¬S	0.1

Calculate:

P(H), P(R|H), P(S|H), P(W|H), P(R|W,H), P(S|W,H)

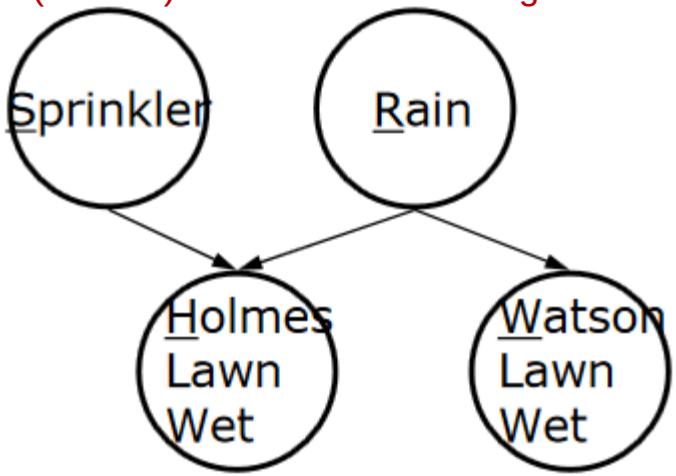
House Lawn Wet. Explained Away!

Holmes and Watson in LA

One morning Holms wakes up to find his lawn wet. He wonders if it has rained or if he left his sprinkler on. He looks at his neighbor Watson's lawn and he sees it is wet too. So, he concludes it must have rained.

Holmes and Watson have moved to LA.

At this time, Prob(Sprinkler=1) and Prob(Rain=1) both increased:
Normally P(Srinkler=1) = 0.1. After observing Holmes Lawn is Wet, P(Sprinkler=1) increases to 0.3382.
Normally P(Rain=1) = 0.2. After observing Holmes Lawn is Wet, P(Rain=1) increases to 0.7353.



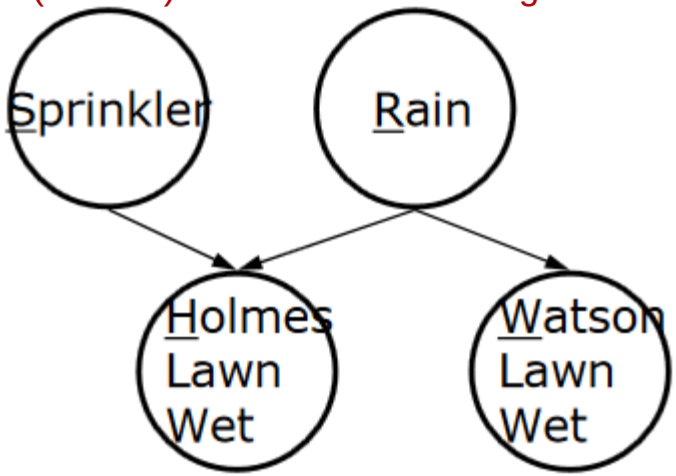
Probability distribution: P(S,R,H,W) = P(H|S,R) P(W|R) P(S) P(R)

House Lawn Wet. The worry is explained away!

Holmes and Watson in LA

One morning Holms wakes up to find his lawn wet. He wonders if it has rained or if he left his sprinkler on. He looks at his neighbor Watson's lawn and he sees it is wet too. So, he concludes it must have rained.

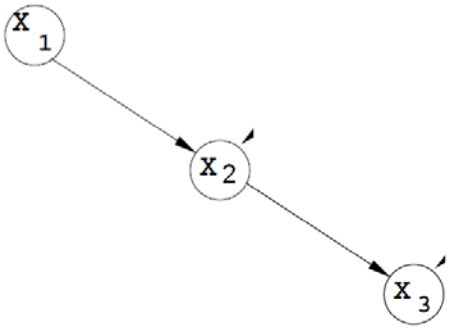
At this time, Prob(Sprinkler=1) and Prob(Rain=1) both increased:
Normally P(Srinkler=1) = 0.1. After observing Holmes Lawn is Wet, P(Sprinkler=1) increases to 0.3382.
Normally P(Rain=1) = 0.2. After observing Watson Lawn is Wet, P(Rain=1) increases to 0.7353.

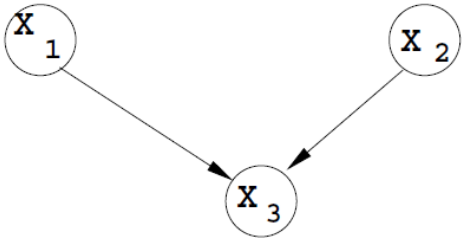


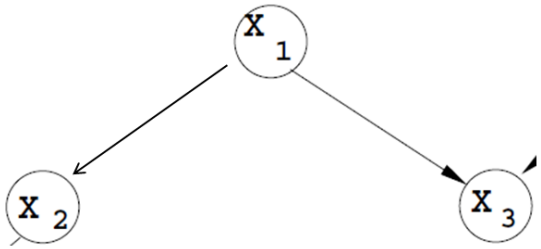
After further observing Watson Lawn is Wet :
P(Rain=1) increased from 0.7353 to 0.9328.
But, P(Sprinkler=1) decreased from 0.3382 to 0.1604

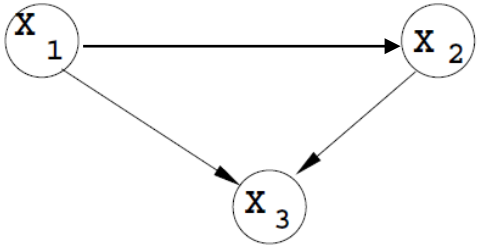
Probability distribution: $P(S,R,H,W) = P(H|S,R)\ P(W|R)\ P(S)\ P(R)$

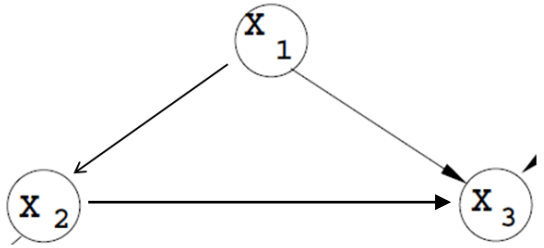
For three random variables (nodes), here are possible network structures



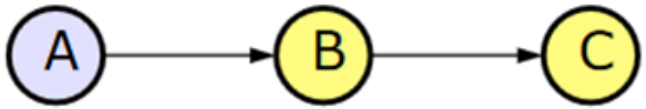








Forward Serial Connection



$$P(A,B,C) = P(C|B)P(B|A)P(A)$$

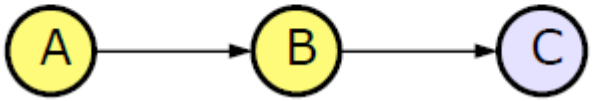
- Knowing about A will tell us something about C
- But, if we know B, then knowing about A will not tell us anything new about C.
- Transmit evidence from A to C through unless B is instantiated (its truth value is known)

$$P(C|A,B) = \frac{P(A,B,C)}{P(A,B)} = \frac{P(C|B)P(B|A)P(A)}{P(B|A)P(A)} = P(C|B)$$

Knowing B renders the knowledge on A irrelevant, i.e. blocking information transfer from A to C.
A is d-separated from C.

A, C becomes conditional independent by fixing B

Backward Serial Connection



$$P(A,B,C) = P(C|B)P(B|A)P(A)$$

- Knowing about C will tell us something about A
- But, if we know B, then knowing about C will not tell us anything new about A

- Transmit evidence from C to A through unless B is instantiated (its truth value is known)

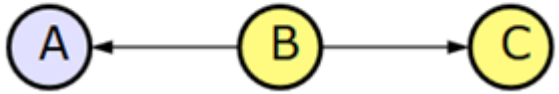
$$P(A|B,C) = \frac{P(A,B,C)}{P(B,C)} = \frac{P(C|B)P(B|A)P(A)}{P(C|B)P(B)} = \frac{P(B|A)P(A)}{P(B)} = P(B|A)$$

Knowing B renders the knowledge on C irrelevant, i.e. blocking information transfer from C to A.

C is d-separated from A

Diverging Connection

Icy road example



$$P(A,B,C) = P(A|B)P(C|B)P(B)$$

When B is open (not fixed, not known)

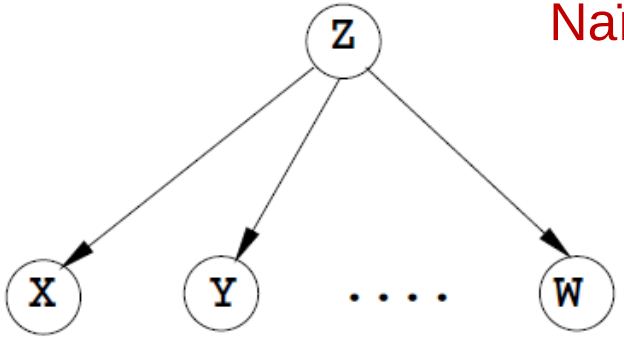
- Knowing about A will tell us something about B, thus something on C
- Knowing about C will tell us something about B, thus something on A

But if B is fixed, knowing A will not help knowing C
Information does not transit thru when B is fixed.

$$P(C|A,B) = \frac{P(A,B,C)}{P(A,B)} = \frac{P(A|B)P(C|B)P(B)}{P(A|B)P(B)} = P(C|B)$$

A and C have a common cause. They are related. But if B is fixed, A,C are independent: d-separated.

Case 3: Diverging connection (common cause)



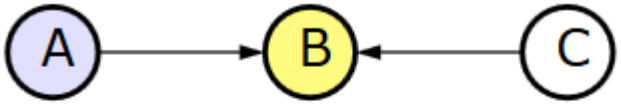
Naïve Bayes Classification

- If Z is not observed, X and Y are dependent.
- **Information can be transmitted through Z among children of Z if Z is not observed.**

- If Z is observed, X and Y are independent.
- **Information cannot be transmitted through Z among children of Z if Z is observed. Observing Z blocks the information path.**

independent. d-separated.

Converging Connection



B is a “colider”

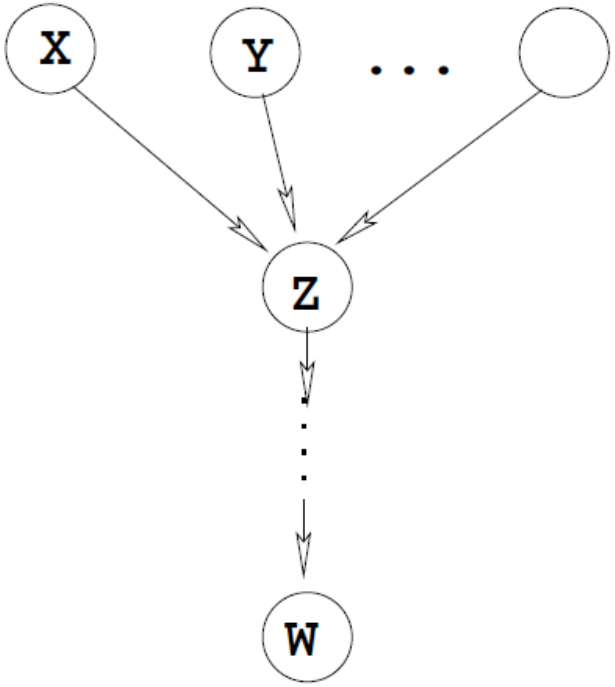
$$P(A,B,C) = P(B|A,C)P(A)P(C)$$

- Transmit evidence from A to C only if B or a descendant of B is instantiated
  - A = Bacterial infection
  - B = Sore throat
  - C = Viral Infection
- Without knowing B, finding A does not tell us anything about C

Reasoning: B could be 0 or 1. We include both cases by summing over B. This makes A,C independent:

$$\text{sum}_B P(A,B,C) = sum_B P(B|A,C)P(A)P(C) = P(A)P(C)$$

Case 4: Converging connection (common effect)



- If neither Z nor any of its descendant are observed, X and Y are independent.
- **Information cannot be transmitted through Z among parents of Z. It leaks down Z and its descendants.**
- If Z or any of its descendant is observed, X and Y are dependent.
- **Information can be transmitted through Z among parents of Z if Z or any of its descendants are observed. Observing Z or its descendants opens the information path.**

- X, Y independent if and only if: $X \perp\!\!\!\perp Y$

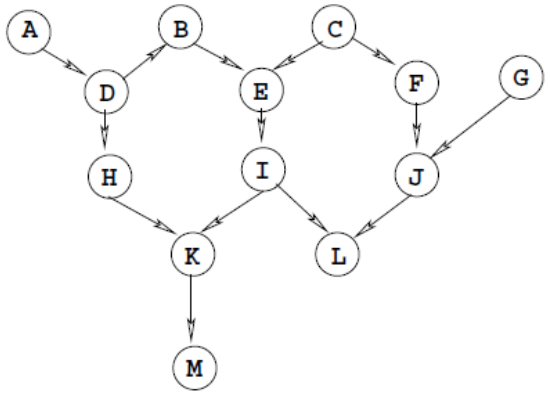
$$\forall x, y : P(x, y) = P(x)P(y)$$

- X and Y are conditionally independent given Z: $X \perp\!\!\!\perp Y | Z$ if and only if:

$$\forall x, y, z : P(x, y|z) = P(x|z)P(y|z)$$

Markov Blanket

- In a Bayesian network, the **Markov blanket** of a node X is the set consisting of
  - Parents of X
  - Children of X
  - Parents of children of X
- Example:



The Markov blanket of I is $\{E, H, J, K, L\}$

In a Bayesian network, varable X is conditionally indepdent of other variables once X's Markov blanket is fixed

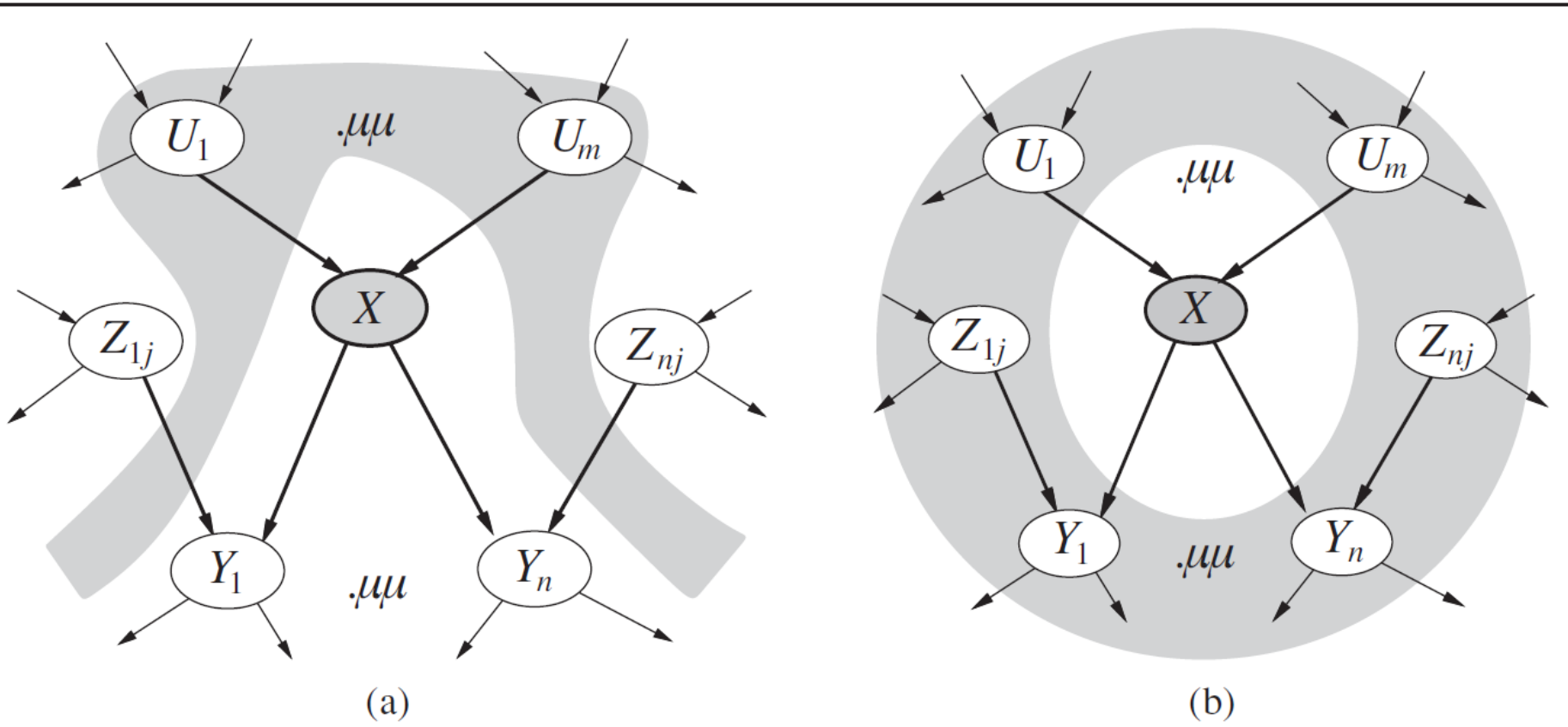


Figure 14.4 (a) A node X is conditionally independent of its non-descendants (e.g., the Z_{ij}s) given its parents (the U_is shown in the gray area). (b) A node X is conditionally independent of all other nodes in the network given its Markov blanket (the gray area).

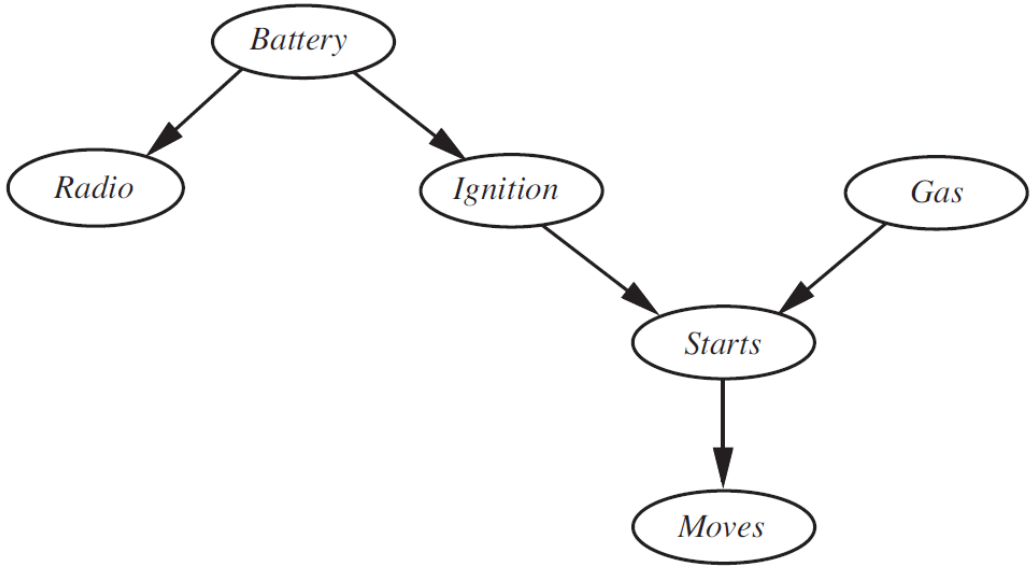


Figure 14.21 A Bayesian network describing some features of a car's electrical system and engine. Each variable is Boolean, and the *true* value indicates that the corresponding aspect of the vehicle is in working order.

14.8 Consider the network for car diagnosis shown in Figure 14.21.

a. Extend the network with the Boolean variables *IcyWeather* and *StarterMotor*.

b. Give reasonable conditional probability tables for all the nodes.

c. How many independent values are contained in the joint probability distribution for eight Boolean nodes, assuming that no conditional independence relations are known to hold among them?

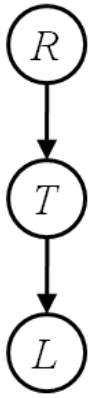
d. How many independent probability values do your network tables contain?

e. The conditional distribution for *Starts* could be described as a **noisy-AND** distribution. Define this family in general and relate it to the noisy-OR distribution.

Compute marginal distribution

using variable elimination:

1. Faster
2. Gives better understanding



$$P(L) = ?$$

- Inference by Enumeration

$$= \sum_t \sum_r P(L|t)P(r)P(t|r)$$

Compute:
(4mult+1add)*2=10

- Variable Elimination

$$= \sum_t P(L|t) \sum_r P(r)P(t|r)$$

Join on r

Eliminate r

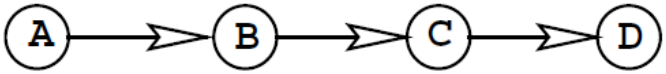
Join on t

Eliminate t

Compute:
(2mult+1add)*2=6

variable elimination (factorize+store intermediate results) speeds up the computation.

- Network: $P(A)$, $P(B|A)$, $P(C|B)$, $P(D|C)$.



- Query: P(D)?
- Computation:

$$P(D) = \sum_{A,B,C} P(A,B,C,D)$$

(6mult+1add)*2*2*2=56

$$= \sum_{C}\sum_{B}\sum_{A} P(A)P(B|A)P(C|B)P(D|C) \quad (1)$$

$$= \sum_{C}\sum_{B} P(C|B)P(D|C)\sum_{A} P(A)P(B|A)$$

$$= \sum_{C} P(D|C) \sum_{B} P(C|B) \sum_{A} P(A)P(B|A) \quad (2)$$

(2mult+1add)*3=9

Homework:

In the Watson and Holmes visit LA network, use the following conditional probability tables.

P(R) = 0.2

P(S) = 0.1

	P(W| R)
R	1.0
¬R	0.2

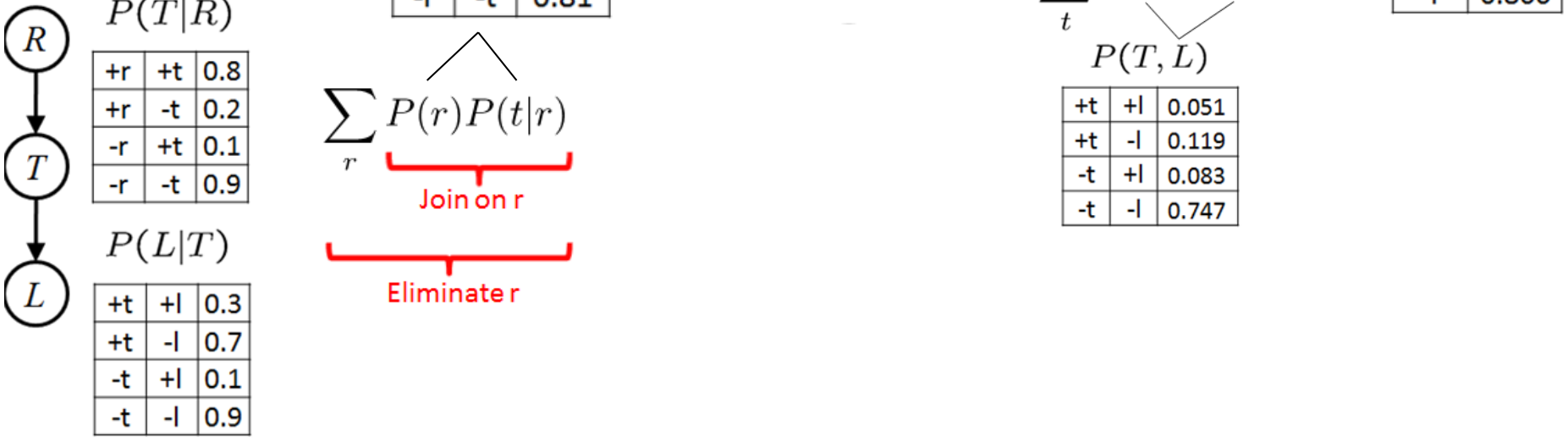
	P(H| R,S)
R,S	1.0
R,¬S	1.0
¬ R,S	0.9
¬ R,¬S	0.1

Calculate:

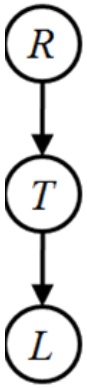
P(H), P(R|H), P(S|H), P(W|H), P(R|W,H), P(S|W,H)

Compute marginal distribution

using variable elimination



Compute marginal distribution



$P(R)$

+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

What is $P(T)$ =?

L is integrated out (marginalized out).
Thus we have:

Join R

$P(R,T)$

+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81

Sum out R

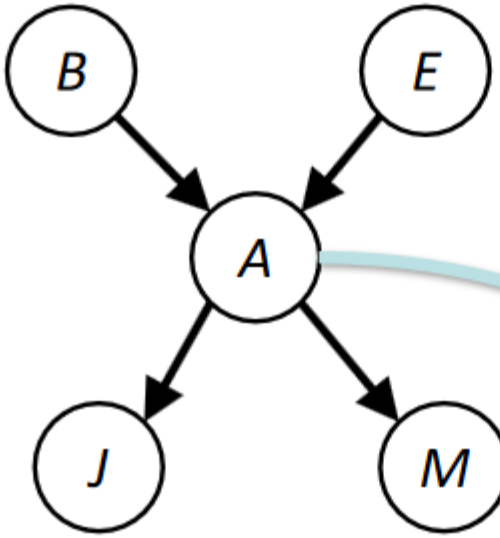
$P(T)$

+t	0.17
-t	0.83

$$\sum_r P(r)P(t|r)$$

Join on r

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

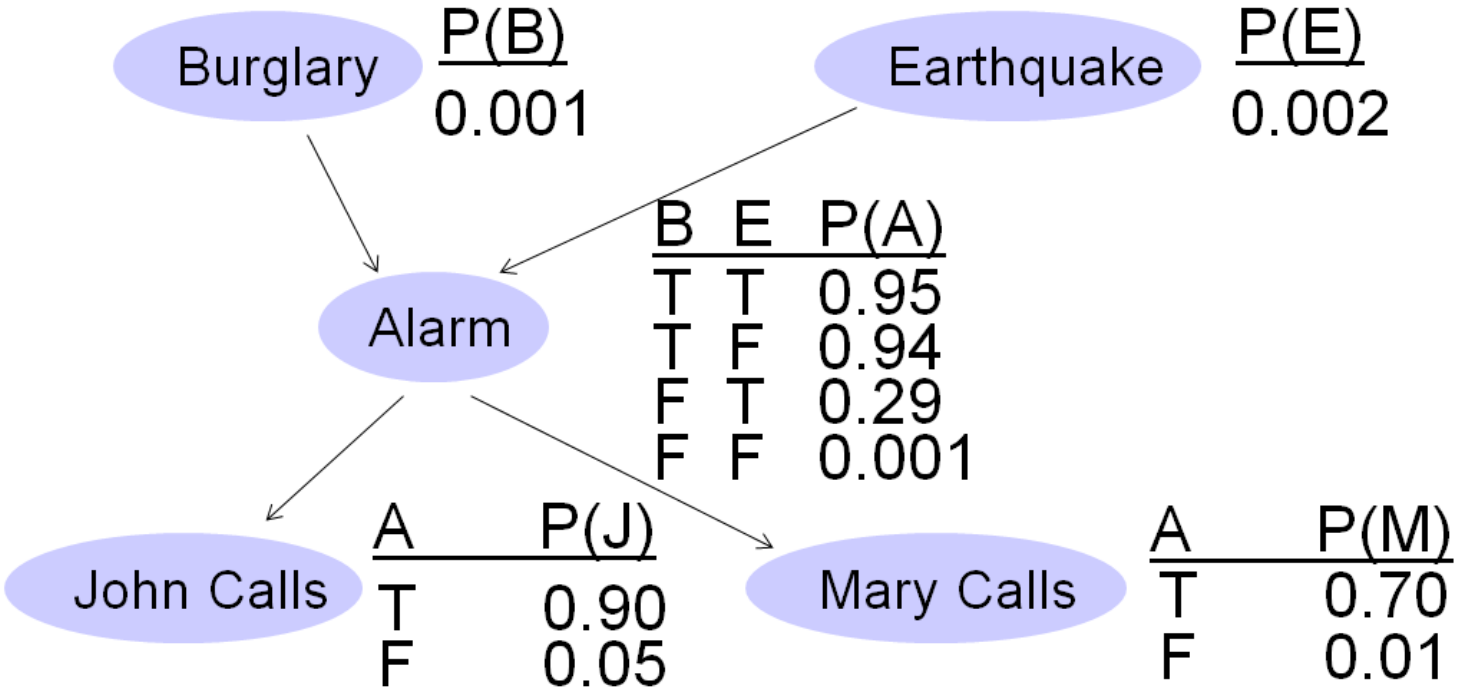


B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

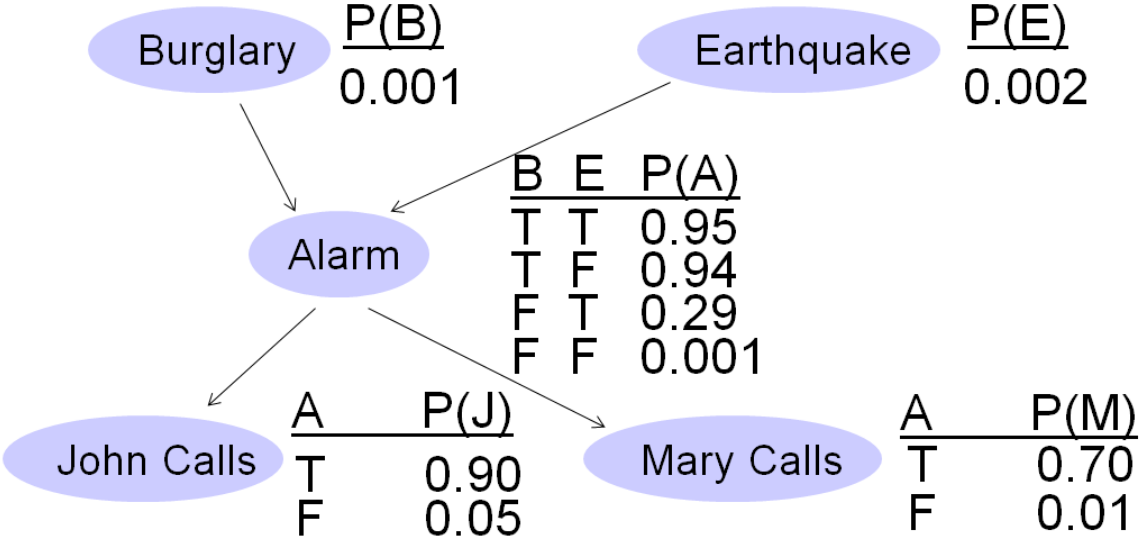
A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

Burglary Alarm Example



$$P(B, E, A, J, M) = P(J|A)P(M|A)P(A|B, E)P(B)P(E)$$

Belief Network



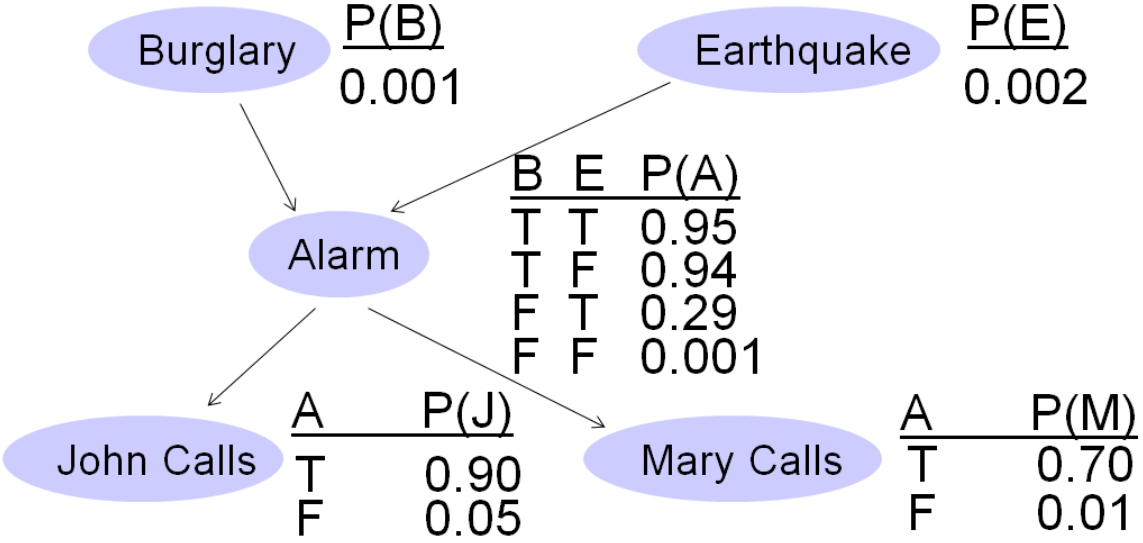
$$P(x_1, x_2, \ldots x_n) = \prod_{i=1}^{n} P(x_i | Parents(X_i))$$

Probability of alarm, no burglary or earthquake, both John and Mary call:

$$P(J|A)P(M|A)P(A|\neg B \wedge \neg E)P(\neg B)P(\neg E)$$

$$(0.9)(0.7)(0.001)(0.999)(0.998) = 0.00062$$

Homework

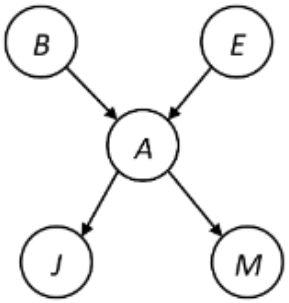


What is Probability of
(burglary happened, earthquake happened,
but John did not call, Mary did not call)

Variable Elimination

$$P(B|j,m) \propto P(B,j,m)$$

$P(B)$	$P(E)$	$P(A|B,E)$	$P(j|A)$	$P(m|A)$



$$\begin{aligned}
P(B|j,m) &\propto P(B,j,m) \\
&= \sum_{e,a} P(B,j,m,e,a) && \text{marginal can be obtained from joint by summing out} \\
&= \sum_{e,a} P(B)P(e)P(a|B,e)P(j|a)P(m|a) && \text{use Bayes' net joint distribution expression} \\
&= \sum_{e} P(B)P(e)\sum_{a} P(a|B,e)P(j|a)P(m|a) && \text{use xy + xz= x*(y+z)} \ \textbf{\textit{do sum first}} \\
&= \sum_{e} P(B)P(e)f_1(B,e,j,m) && \text{joining on a, and then summing out gives } f_1 \\
&= P(B)\sum_{e} P(e)f_1(B,e,j,m) && \text{use xy + xz = x*(y+z)} \ \textbf{\textit{do sum first}} \\
&= P(B)f_2(B,j,m) && \text{joining on e, and then summing out gives } f_2
\end{aligned}$$

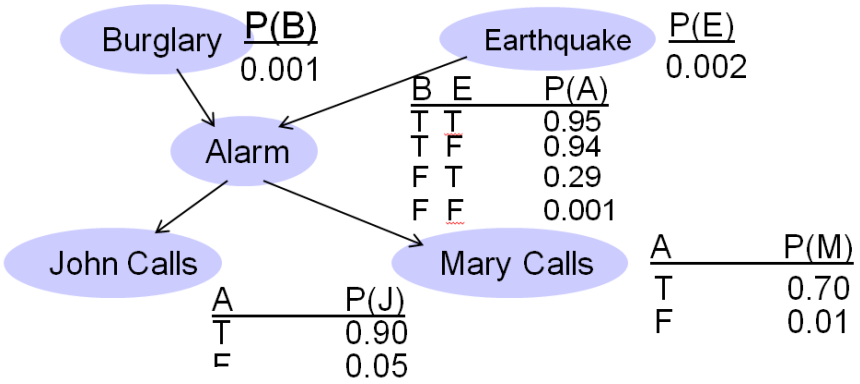
Simple! Exploiting uwy + uwz + uxy + uxz + vwy + vwz + vxy +vxz = (u+v)(w+x)(y+z) to reduce computation

Causal Inferences

Inference from cause to effect. Top down.

A burglary happened. What is the probability John will call ?

Need to compute P(J|B)



$$P(J|B) = P(J,B)/P(B),$$

$$P(J,B) = \sum_{A,E,M} P(J,M,A,B,E) = \sum_{A,E,M} P(J|A)P(M|A)P(A|B,E)P(B)P(E)$$

$$= P(B)\big[\sum_{M} P(M|A)\big] \sum_{A} P(J|A) \sum_{E} P(A|B,E)P(E),$$

$$= P(B) \sum_{A} P(J|A) \sum_{E} P(A|B,E)P(E),$$

$P(A|B) = \sum_{E} P(A|B,E)P(E)$ is conditional probability

Use short-hands notations $P(A^1|B^1) \triangleq P(A=1|B=1), P(A^0|B^1) \triangleq P(A=0|B=1),$

$$P(A^1|B^1) = (0.95)(0.002) + (0.94)(1-0.002) = 0.94002,$$

$$P(A^0|B^1) = 1 - P(A^1|B^1) = 0.05998$$

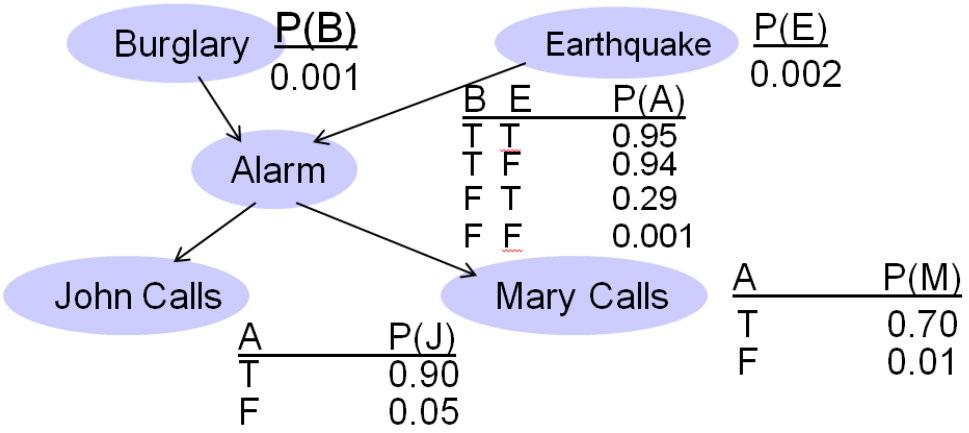
$$P(J^1,B^1) = P(B^1)\big[(0.9)(0.94002) + (1-0.9)(0.05998)\big] = P(B^1)(0.852016)$$

$$P(J^1|B^1) = P(J^1,B^1)/P(B^1) = 0.852016$$

Causal Inferences

Inference from cause to effect.

Given a burglary, what is P(J|B)?



$$P(J|B) = P(J,B)/P(B),$$

$$P(J,B) = \sum_{A,E,M} P(J,M,A,B,E) = \sum_{A,E,M} P(J|A)P(M|A)P(A|B,E)P(B)P(E)$$

$$= P(B)\big[\sum_{M} P(M|A)\big] \sum_{A} P(J|A) \sum_{E} P(A|B,E)P(E),$$

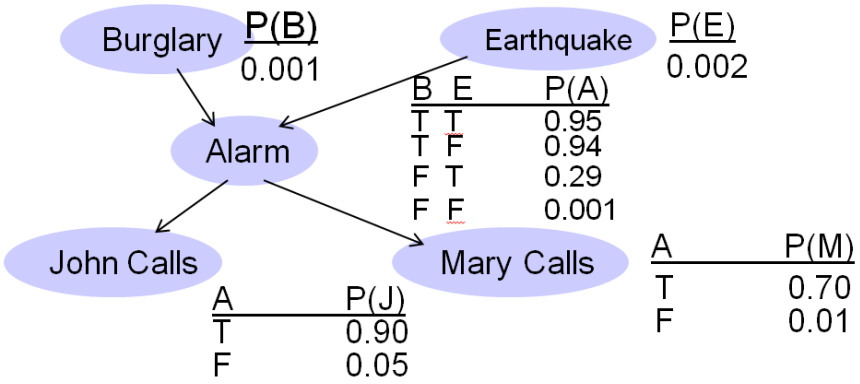
$$= P(B) \sum_{A} P(J|A) \sum_{E} P(A|B,E)P(E),$$

Computing conditional probability in Bayesian Network

$$P(J|B) = P(J,B)/P(B),$$

$$\begin{aligned} P(J,B) &= \sum_{A,E,M} P(J,M,A,B,E) = \sum_{A,E,M} P(J|A)P(M|A)P(A|B,E)P(B)P(E) \\ &= P(B)\big[\sum_{M} P(M|A)\big] \sum_{A} P(J|A) \sum_{E} P(A|B,E)P(E), \\ &= P(B) \sum_{A} P(J|A) \sum_{E} P(A|B,E)P(E), \end{aligned}$$

$$P(A|B) = \sum_{E} P(A|B,E)P(E),$$ is conditional probability



Use short-hand notations:

$$P(A^1|B^1) \triangleq P(A=1|B=1), P(A^0|B^1) \triangleq P(A=0|B=1),$$

$$\begin{aligned} P(A^1|B^1) &= (0.95)(0.002) + (0.94)(1-0.002) = 0.94002, \\ P(A^0|B^1) &= 1 - P(A^1|B^1) = 0.05998 \end{aligned}$$

$$P(J^1,B^1) = P(B^1)\big[(0.9)(0.94002) + (1-0.9)(0.05998)\big] = P(B^1)(0.852016)$$

$$P(J^1|B^1) = P(J^1,B^1)/P(B^1) = 0.852016$$

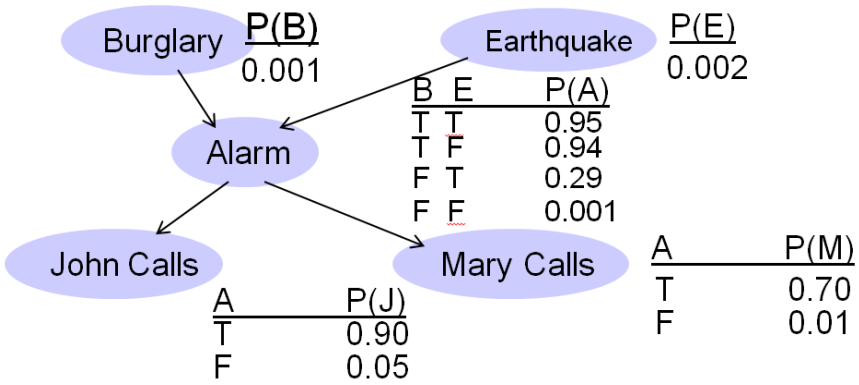
Diagnostic Inferences

From effect to cause (bottom up)

John called, what is P(burglary)?

$$P(B \mid J) = \frac{P(J \mid B)P(B)}{P(J)}$$

What is P(J)? Need P(A) first:



$$P(B,E,A,J,M) = P(J|A)P(M|A)P(A|B,E)P(B)P(E)$$

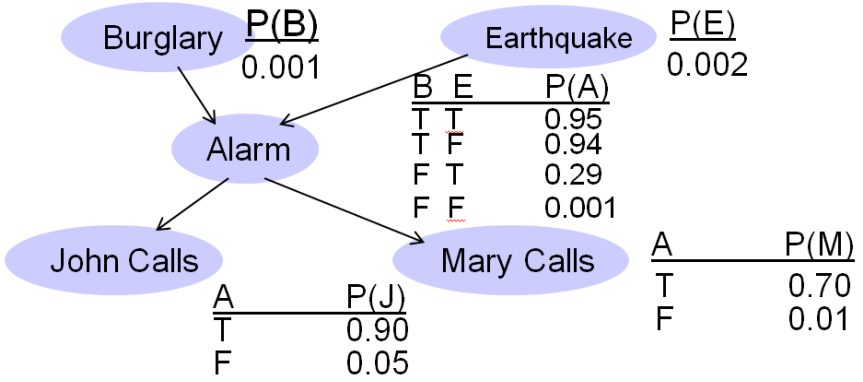
$$P(A) = P(B)P(E)(0.95) + P(\neg B)P(E)(0.29)$$
$$+ P(B)P(\neg E)(0.94) + P(\neg B)P(\neg E)(0.001)$$
$$P(A) = (0.001)(0.002)(0.95) + (0.999)(0.002)(0.29)$$
$$+ (0.001)(0.998)(0.94) + (0.998)(0.999)(0.001)$$
$$P(A) = 0.002517$$

$$P(J) = P(A)(0.9) + P(\neg A)(0.05)$$
$$P(J) = (0.002517)(0.9) + (0.9975)(0.05)$$
$$P(J) = 0.052$$

$$P(B \mid J) = \frac{(0.85)(0.001)}{(0.052)} = 0.016$$

More complex Inferences

John called and burglary happened, what is Prob(Alarm)?



P(A|B,J) = P(A,B,J)/P(B,J)

$$P(B,E,A,J,M) = P(J|A)P(M|A)P(A|B,E)P(B)P(E)$$

P(A,B,J) = sum_E P(A,B,J,E) = sum_E P(A|B,E)P(B)P(E) P(J|A)

P(A=1,B=1,J=1) = [sum_E P(A=1|B=1,E=1)P(B=1)P(E)]P(J=1|A=1)
= [0.95(0.001)(0.002)+0.94(0.001)(0.998)](0.90) = 84.6*10^(-5)

P(A=0,B=1,J=1) = [sum_E P(A=0|B=1,E=1)P(B=1)P(E)]P(J=1|A=0)
= [(1-0.95)(0.001)(0.002)+(1-0.94)(0.001)(0.998)](0.05) = 3.1*10^(-5)

P(A=1|B=1,J=1) = P(A=1,B=1,J=1)/P(B=1,J=1) = 84.6/(84.6+3.1)=0.965

HOMEWORK: John called but no burglary happened, what is Prob(Alarm)?

So far, we discussed the DAG (Bayesian networks).

When the graph has undirected edges, what would happen?

Markov Random Field

MRF Definition

A Markov Random Field is determined by

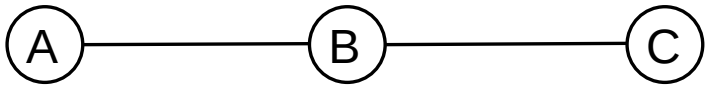
- a set of sites $\mathcal{S} = \{1 \dots N\}$. These will correspond to the N pixel locations,
- a set of random variables $y = \{y_1 \dots y_N\}$ associated with each of the sites,
- a set of neighbors $\mathcal{N}_{1\dots N}$ at each of the N sites. The set $\mathcal{N}_n$ contains the indices of the subset of random variables have an immediate probabilistic connection to variable y_n.

To be a Markov random field, the model must obey the Markov property,

$$Pr(y_n|y_{\mathcal{S}\setminus n}) = Pr(y_n|y_{\mathcal{N}_n}) \qquad \forall \quad n \in \mathcal{S},$$

Bayesian Network is a directed acyclic graph

Markov Random Fields undirected graph (could have cycles)

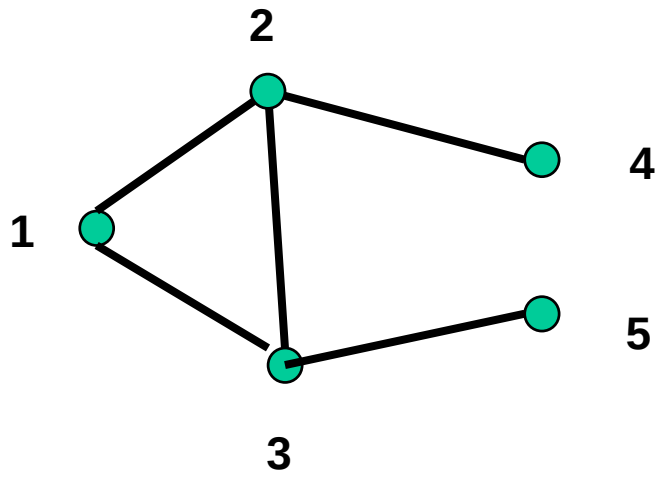


A is independent of C given B

Random markov fields has wide applications in computer vision,
text mining,
natural language processing,
computational biology
etc

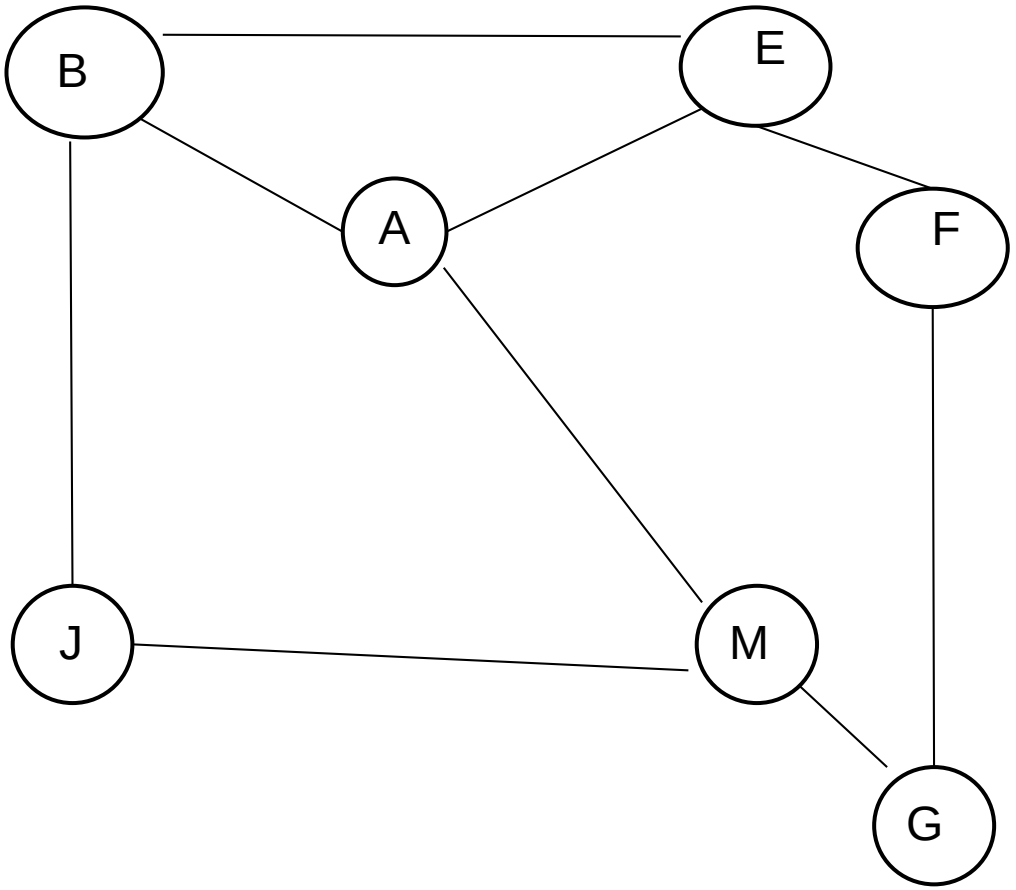
Random Markov Fields

Undirected edges connect nodes.



Given node 2, node1 and node 4 are independent.

{A,B,J} indep { F, G} given {E, M}



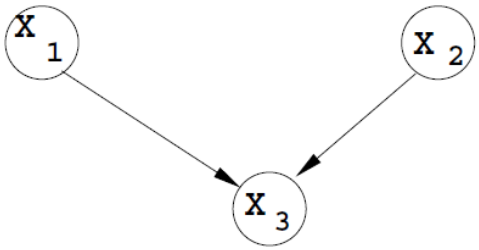
P(A,B,J| E, M)

Bayesian net structures and parameters estimation using MLE

Given data below, compute probabilities for Bayesian network:

Compute model parameters, data likelihood

X_1	X_2	X_3
1	1	1
1	1	2
1	1	2
1	2	2
1	2	2
1	2	2
2	1	1
2	1	1
2	1	1
2	1	2
2	2	1
2	2	1
2	2	2
2	2	2
2	2	2
2	2	2



$P(X_1=1)$ is: $6/16$

$P(X_2=1)$ is: $7/16$

$P(X_3=1|X_1=2, X_2=2)$ is: $2/6$

P(x3=1|x1=1,x2=1) =1/3
P(x3=1|x1=1,x2=2) = 0
P(x3=1|x1=2,x2=1) = ¾

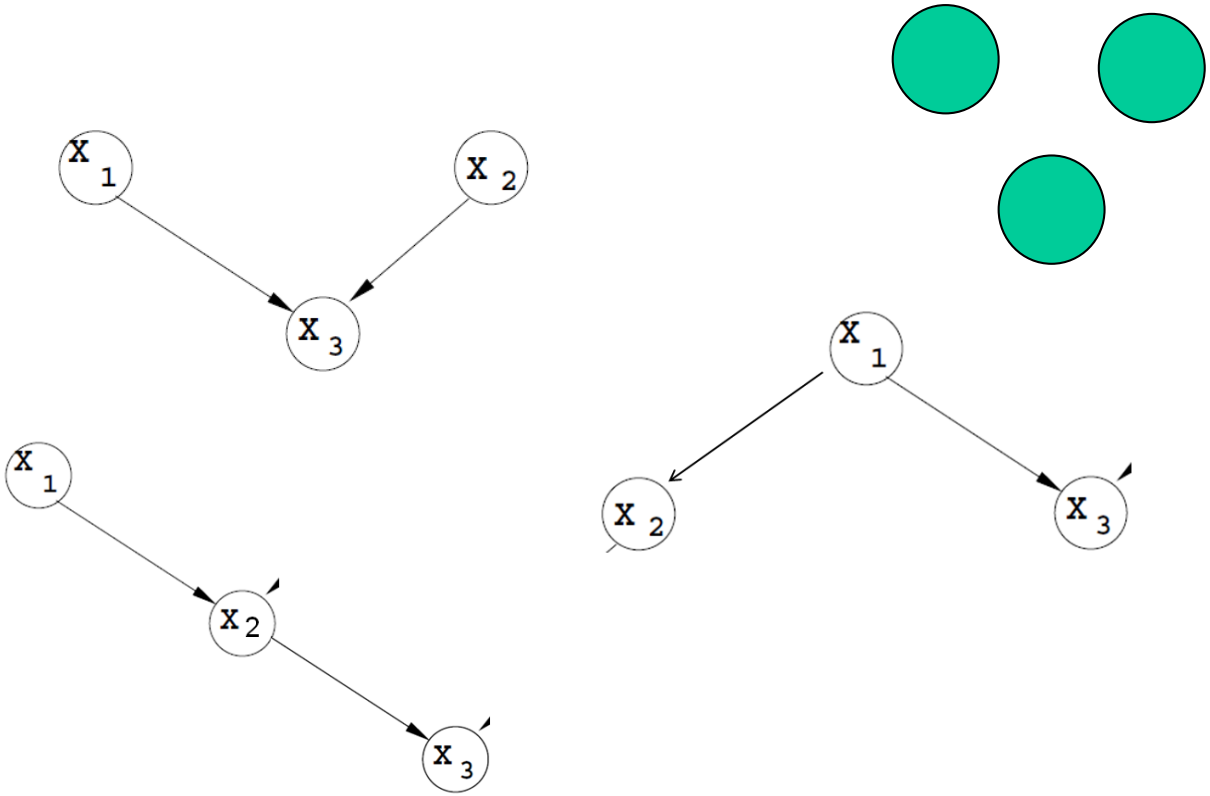
Likelihood = P1 *P2 * P3 *… P*16
= (6/16 * 7/16 * 1/3) * (6/16 * 7/16 * 2/3)
*(10/16 * 9/16 * 4/6)

Given data below.

Which Bayesian network structure is correct?

X_1	X_2	X_3
1	1	1
1	1	2
1	1	2
1	2	2
1	2	2
1	2	2
2	1	1
2	1	1
2	1	1
2	1	2
2	2	1
2	2	1
2	2	2
2	2	2
2	2	2
2	2	2

Simplest one is assume 3 variables are independent

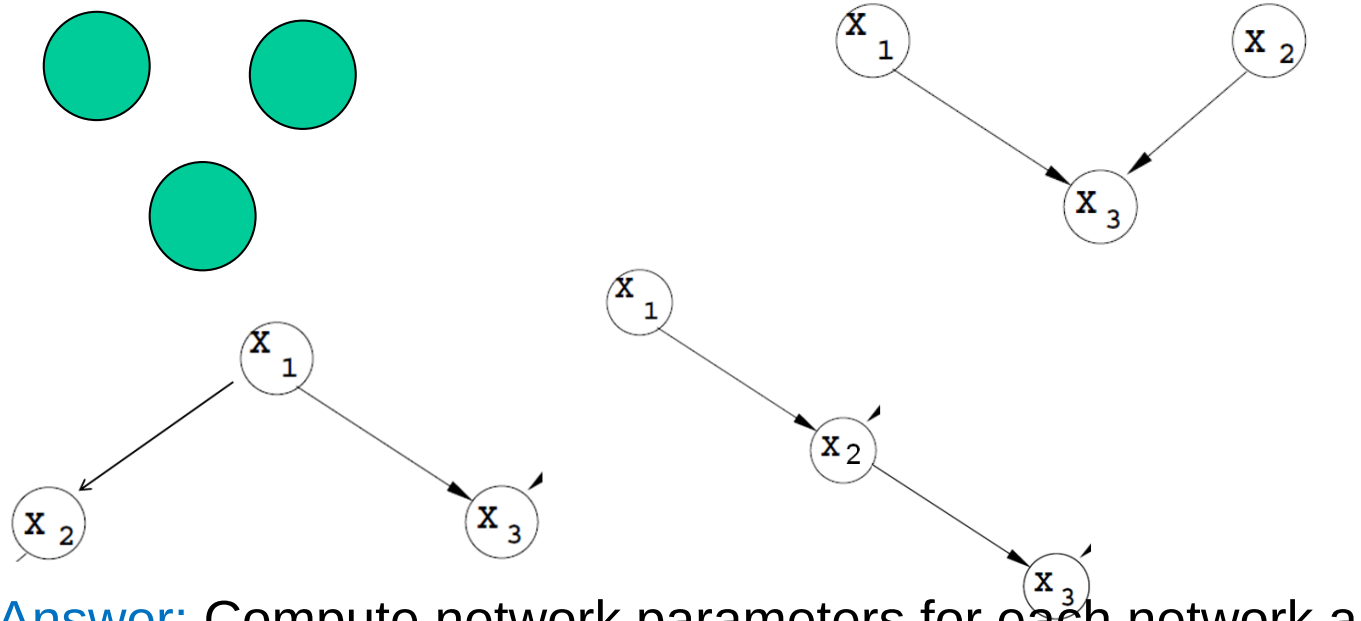


Answer: Compute network parameters for each network and their corresponding probability likelihood. Chose the one with maximum likelihood. But ……

Given data below.

Which Bayesian network structure is correct?

X_1	X_2	X_3
1	1	1
1	1	2
1	1	2
1	2	2
1	2	2
1	2	2
2	1	1
2	1	1
2	1	1
2	1	2
2	2	1
2	2	1
2	2	2
2	2	2
2	2	2
2	2	2



Answer: Compute network parameters for each network and their corresponding probability likelihood. Chose the one with maximum likelihood.

But complex models with more parameters usually fits data better and thus gives higher likelihood. Simply using MLE will choose these complex models.

Generally we prefer simpler models because they have better generalization.

Model selection with different penalties/regularization are proposed.

Homework.







Day: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Weather: S S R C C R R S S S R C C R S S C

Given above sequence of 17 weather events, estimate the transition probability A:

$$A_{x_{t-1},x_t} = P(x_t \mid x_{t-1})$$

Hint: Draw contingency table. Compute conditional probabilities.

Thursday, March 26, No class

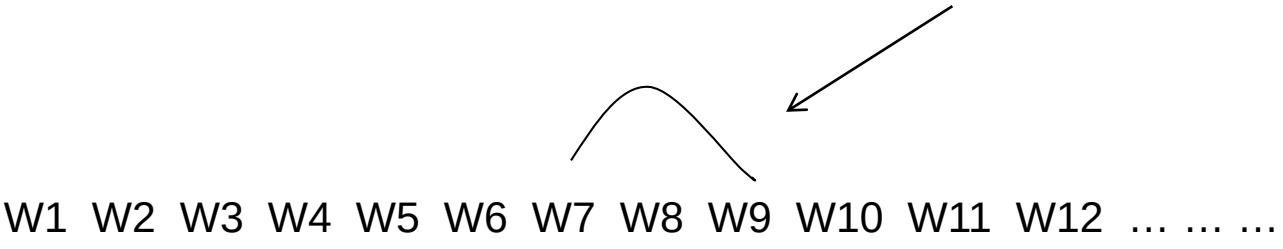
Tuesday March 31, A brief review/highlights of covered lectures

Homework

What is the probability of 3 contiuous sunny day ?
Assume weather={**S**unny, C**l**oudy , **R**aining} with transition probability matrix:

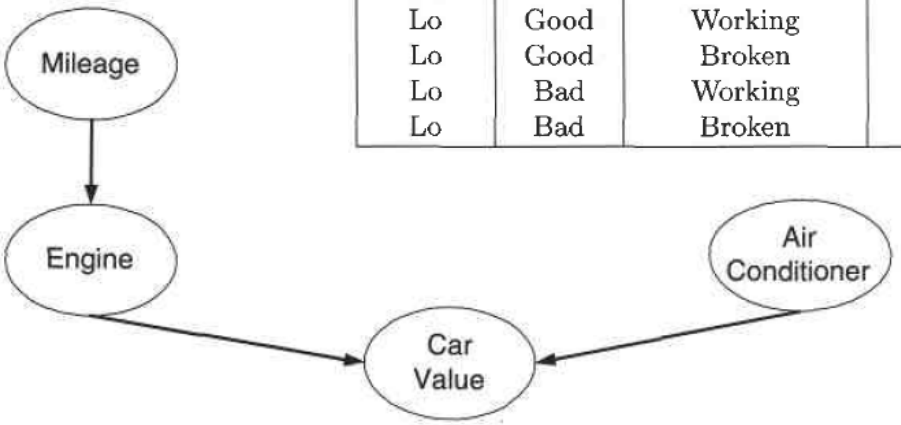
$$A = \begin{pmatrix} .8 & .15 & .05 \\ .38 & .6 & .02 \\ .75 & .05 & .2 \end{pmatrix}$$

3-day sequence can start any day



Homework:

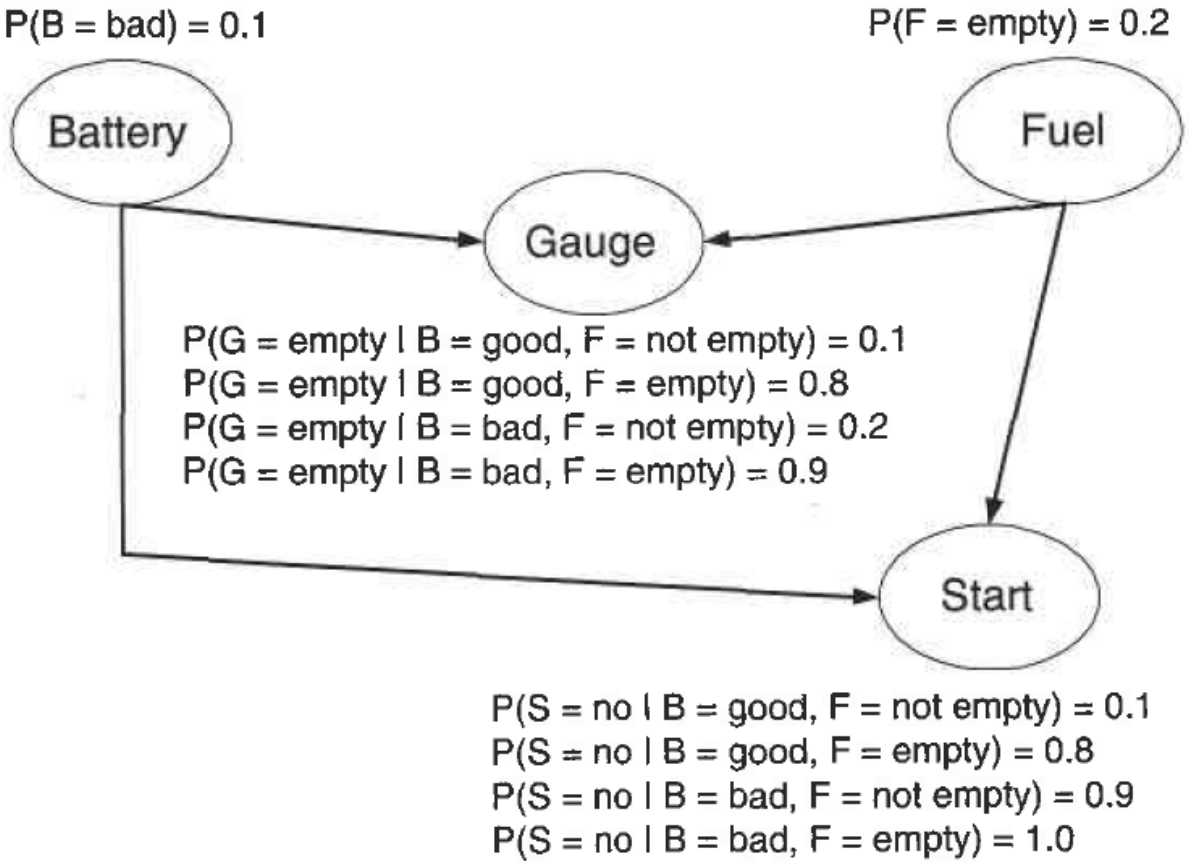
Mileage	Engine	Air Conditioner	Number of Records with Car Value=Hi	Number of Records with Car Value=Lo
Hi	Good	Working	3	4
Hi	Good	Broken	1	2
Hi	Bad	Working	1	5
Hi	Bad	Broken	0	4
Lo	Good	Working	9	0
Lo	Good	Broken	5	1
Lo	Bad	Working	1	2
Lo	Bad	Broken	0	2



This line means there are 3+4 data instances:
3 instances for car-value= High,
4 instances for car-value = Low

Given Bayesian Network, and the training data instances.
Tasks:
1. Write down the probability distribution decomposition.
2. Learn network parameters: P(M), P(A), P(E|M), P(C|E,A).
   Hint: When estimate P(C|E,A), you need to integrate out *M.* Thus there are 3+4+9+0
   - data instances with E=good AND A=working , 3+9 instances for C=Hi, and
   - 4+0 instances for C=low.
3. Compute Prob(Engine = bad, Air Condition = Broken)

Homework Problem 4.



Tasks:

1. Write down the probability distribution decomposition.
3. Compute:

(a) $P(\mathrm{B} = \text{good}, \mathrm{F} = \text{empty}, \mathrm{G} = \text{empty}, \mathrm{S} = \text{yes})$.

(b) $P(\mathrm{B} = \text{bad}, \mathrm{F} = \text{empty}, \mathrm{G} = \text{not empty}, \mathrm{S} = \text{no})$.

(c) Given that the battery is bad, compute the probability that the car will start.